ICMA424

Abstract Algebra

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Trimester 1, 2023-2024

Contents

0 Preliminaries **3**
- 0.1 Properties of Integers . . . 3
  - 0.1.1 The Well Ordering Principle and Mathematical Induction . . . 3
  - 0.1.2 Divisibility . . . 5
- 0.2 Equivalence Relations . . . 8

1 Introduction to Groups **11**
- 1.1 Symmetries of a Square . . . 11
- 1.2 The Dihedral Groups . . . 14

2 Groups **15**
- 2.1 Definition and Examples . . . 15
- 2.2 Elementary Properties of Groups . . . 16
- 2.3 $\mathbb{Z}_n$ and $U(n)$ . . . 18
- 2.4 Bézout's Lemma . . . 18

3 Finite Groups and Subgroups **19**
- 3.1 Terminology and Notation . . . 19
- 3.2 Subgroup Tests . . . 21
- 3.3 Examples of Subgroups . . . 23

4 Cyclic Groups **27**
- 4.1 Classification of Subgroups of Cyclic Groups . . . 33

5 Permutation Groups **35**
- 5.1 Cycle Notation . . . 37
- 5.2 Properties of Permutations . . . 38

6 Isomorphisms **43**
- 6.1 Motivation . . . 43
- 6.2 Definition and Examples . . . 44
- 6.3 Cayley's Theorem . . . 45
- 6.4 Properties of Isomorphisms . . . 46
- 6.5 Automorphisms . . . 47

7 Cosets and Lagrange's Theorem **50**
- 7.1 Properties of Cosets . . . 50
- 7.2 Lagrange's Theorem . . . 53
- 7.3 The Sylow Theorems . . . 56

8 External Direct Products **57**
- 8.1 Definition and Examples . . . 57
- 8.2 Properties of External Direct Products . . . 58
- 8.3 The Group of Units Modulo **n** as External Direct Products . . . 60

9 Normal Subgroups and Factor Groups **62**
- 9.1 Normal Subgroups . . . 62
- 9.2 Factor Groups . . . 63
- 9.3 Applications of Factor Groups . . . 64
- 9.4 Internal Direct Products . . . 66

10 Group Homomorphisms **68**
10.1 Properties of Homomorphisms 69
10.2 The First Isomorphism Theorem 72

11 Fundamental Theorem of Finite Abelian Groups **74**
11.1 The Fundamental Theorem 74

12 Introduction to Rings **75**
12.1 Motivation and Definition 75
12.2 Examples of Rings 76
12.3 Properties of Rings 76
12.4 Subrings 78

13 Integral Domains **79**
13.1 Definition and Examples 79
13.2 Fields 80

Chapter 0

Preliminaries

Let's begin by clarifying how algebra differs from arithmetic. Arithmetic is all about dealing with *numbers* and their operations directly. Algebra aims to explore the *structure* of numbers. About 1200 years ago, the philosopher/astronomer Musa al-Khwarizmi* introduced algebra (then called *al-jebr*) in one of his textbooks.

Figure 1: Musa al-Khwarizmi [Source: University of Waterloo]

What about *abstract* algebra? In this context, the word "abstract" means *structured* or *simplified*, i.e., abstract algebra attempts to simplify the structure of algebra.

0.1 Properties of Integers

0.1.1 The Well Ordering Principle and Mathematical Induction

One important property of integers that we will often use is the *Well-Ordering Principle*. We will treat it as an axiom, as this property cannot be proved using the usual properties of arithmetic.

*The word *algorithm* comes from his name as well.

Well Ordering Principle. *Every nonempty set of positive integers contains a smallest member.*

One application of the Well Ordering Principle is the Principle of Mathematical Induction, or rather, the First Principle of Mathematical Induction, which states that for every $n \in \mathbb{Z}^+$, if $T(n)$ is a statement and

(1) $T(1)$ is true and

(2) $T(k)$ implies $T(k+1)$, $\forall k \in \mathbb{Z}^+$.

then $T(n)$ is true for all $n \in \mathbb{Z}^+$. More rigorously

Theorem 0.5 (First Principle of Mathematical Induction). Let S be a set of integers containing a. Suppose S has the property that whenever some integer $n \geq a$ belongs to S, then the integer $n+1$ also belongs to S. Then S contains every integer greater than or equal to a.

Proof. Assume that $T(n)$ satisfies (1) and (2) for each $n \in \mathbb{Z}^+$. Define the set S to be

$$S = \{n \in \mathbb{Z}^+ \mid T(n) \text{ is false } \} \subseteq \mathbb{Z}^+.$$

In the case that $S = \emptyset$, we are done. Assume to the contrary that S is non-empty. By the Well Ordering Principle, S has the smallest member, say $a \in S$. By (1), we know that $T(1)$ is true, thus $1 \notin S$. Thus, $a > 1$. Let $k = a - 1 \in \mathbb{Z}^+$. Note that $k \notin S$, as a is the smallest member of S, therefore $T(k)$ is true. By (2), we have that $T(k+1) = T(a)$ must be true. This is a contradiction, since as $a \in S$, it must be the case that $T(a)$ is false. □

There is also a slightly different version known as the Second Principle of Mathematical Induction[†], which states that for each $n \in \mathbb{Z}^+$, if $T(n)$ is a statement and

(1) $T(1)$ is true and

(2) $\forall k \in \mathbb{Z}^+, T(1), T(2), \ldots, T(k) \implies T(k+1)$,

we have that $T(n)$ is true for all $n \in \mathbb{Z}^+$. More formally

Theorem 0.6 (Second Principle of Mathematical Induction). Let S be a set of integers containing a. Suppose S has the property that n belongs to S whenever every integer less than n and greater than or equal to a belongs to S. Then, S contains every integer greater than or equal to a.

The proof for this can be done using the Well Ordering Principle or The First Principle of Mathematical Induction. Induction itself and its proof were introduced quite recently by Giovanni Vacca in the early 1900s.

Example 1. *Tower of Hanoi.*

As an example of mathematical induction, let us (again) examine the Tower of Hanoi, which involves rearranging disks of increasing size on three pegs. We have the constraint that

[†]Also known as strong induction.

larger disks may not be placed on top of smaller disks and wish to find the minimum number of moves required to move a stack of n disks from one peg to another.

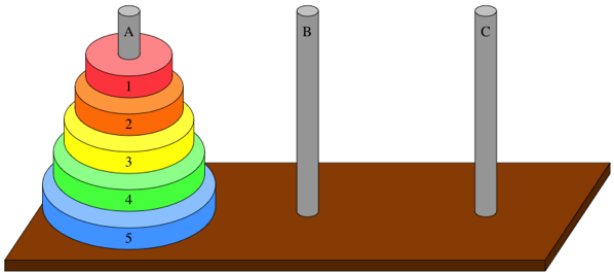


Figure 2: Tower of Hanoi with 5 disks [Source: Khan Academy]

Let $T(n)$ denote the minimum number of steps required to solve Tower of Hanoi. Upon playing a couple games, we see that

$$\begin{aligned} T(1) &= 1 \\ T(2) &= 3 \\ T(3) &= 7 \\ &\vdots \end{aligned}$$

We make an educated guess that $T(n) = 2^n - 1$. Let us prove it using mathematical induction.

Proof. For the base case, $T(1) = 1$, as we can just move the lone disk from its current pole to the target pole. Assume that $T(k)$ holds, i.e., it requires $2^k - 1$ steps to solve a Tower with k disks. We would like to show that the statement holds for a Tower of $k+1$ disks.

Analyzing this, we find that to solve a Tower with $k+1$ disks, we need merely to move the top k disks from their starting pole to an alternate pole, move the bottom largest disk to the target pole, then move the k disks from the alternate pole to the target pole. We know from the base case and inductive hypothesis that these require $2^k - 1$, 1, and $2^k - 1$ steps, respectively. Thus

$$\begin{aligned} T(k+1) &= 2T(k) + 1 \\ &= 2(2^k - 1) + 1 \\ &= 2^{k+1} - 1. \end{aligned}$$

Therefore, by the principle of mathematical induction, we may conclude that the minimum number of steps required to solve Tower of Hanoi with n disks is $2^n - 1$. □

0.1.2 Divisibility

Next we will explore divisibility, GCD, primes, the concept of unique factorization. The concept of divisibility plays an important role in the theory of numbers. We introduce its formal definition below

Definition (Divisibility). Suppose $a, b \in \mathbb{Z}$. We say that a divides b, denoted $a \mid b$, if $\exists c \in \mathbb{Z}$ such that $b = ac$. In other terms, a is a divisor of b, b is a multiple of a, or b is divisible by a.

With the concept of divisibility in mind, we can define what it means for certain numbers to be *prime*.

Definition (Primeness). An integer $p > 1$ is prime if for all $a, b \in \mathbb{Z}$, $p = ab$, then $a = \pm 1$ or $b = \pm 1$.

As another application of the Well Ordering Principle, we can establish a fundamental property of integers: *the division algorithm*.

Theorem 0.1 (Division Algorithm). Let a and b be integers with $b > 0$. Then, there exist unique integers q (the *quotient*) and r (the *remainder*) with the property that $a = bq + r$, where $0 \leq r < b$.

Proof. Given than a and b are integers with $b > 0$, we define the set S to be

$$S = \{k \in \mathbb{Z}^+ \cup \{0\} \mid k = a - bn, \exists n \in \mathbb{Z}\}.$$

To apply the Well Ordering Principle, we must show that S is non-empty. For $n = 0$, we have that

$$k = a - b(0) = a \in \mathbb{Z}^+ \cup \{0\},$$

which is in S. By the Well Ordering Principle, S has the smallest member, say $r \in S$. Thus, $r = a - bq \geq 0$ for some $q \in \mathbb{Z}$. We claim that $0 \leq r < b$. Assume for the sake of contradiction that $r \geq b > 0$. Consider $r' = a - (q-1)b$.

$$\begin{aligned} r' &= a - (q+1)b \\ &= a - qb - b \\ &= r - b \geq 0. \end{aligned}$$

Thus, $r' \in S$ but $r' < r$. This is a contradiction, since r is the smallest member of S. We have proved that there exist q and r as required, however, now we must prove that they are a unique pair. Assume that there are $q_1, q_2 \in \mathbb{Z}$ and $r_1, r_2 \in \mathbb{Z}^+ \cup \{0\}$ such that

$$q = bq_1 + r_1 = bq_2 + r_2,$$

where $0 \leq r_1, r_2 < b$. If we can prove that $r_1 = r_2$, it must follow that $q_1 = q_2$. Without loss of generality, assume that $r_2 \geq r_1$. We have that

$$\begin{aligned} bq_1 + r_1 &= bq_2 + r_2 \\ 0 &= bq_2 + r_2 - (bq_1 + r_1) \\ 0 &= b(q_2 - q_1) + (r_2 - r_1) \\ -(r_2 - r_1) &= b(q_2 - q_1). \end{aligned}$$

Keep in mind that

$$0 \leq r_1, r_2 < b \implies 0 \leq r_2 - r_1 < b.$$

Observing that $0 \leq r_2 - r_1 < b$ and $b \mid (r_2 - r_1)$, we can conclude that $r_2 - r_1 = 0$. Since $0 = b(q_2 - q_1)$, it means either $b = 0$ or $q_2 - q_1 = 0$. As $b > 0$, we have that $q_2 - q_1 = 0$. We can conclude then that $q_1 = q_2$ and $r_1 = r_2$. $\square$

Let's take a look at an example of applying the division algorithm.

Example 2. Suppose that $a = 5$ and $b = 2$, we can write $5 = 2 \cdot 2 + 1$, where $0 \leq 1 < 2$. We can also do the same for the reverse. Suppose $a = 2$ and $b = 5$. We can simply set $q = 0$, i.e., $2 = 5(0) + 2$. Note that this satisfies $0 \leq 2 < 5$.

Next we will introduce the *greatest common divisor* and what it means for integers to be *relatively prime*.

Definition (Greatest Common Divisor/Relative Primes). The greatest common divisor of two nonzero integers a and b is the largest of all common divisors of a and b. We denote this integer by $\gcd(a,b)$. When $\gcd(a,b) = 1$, we say a and b are relatively prime.

To find the greatest common divisor of two integers, we can *Euclid's Algorithm*, which works by repeatedly applying the division algorithm.

Theorem (Euclidean Algorithm). Let $r_0 = a$ and $r_1 = b$ be integers with $a > b > 0$. We apply the division algorithm

$$x = yq + r, \quad 0 \leq r < y$$

to obtain

$$r_i = r_{i+1}q_{i+1} + r_{i+2}$$

with $0 < r_{i+2} < r_{i+1}$ for $0 \leq i < n-1$ and $r_{n+1} = 0$.

We can prove that this algorithm always returns an answer in a finite number of steps using the Well Ordering Principle but we will not do so here. We can show how this works with an example.

Example 3. We will use the Euclidean algorithm to find $\gcd(234, 42)$.

$$\begin{aligned} 234 &= 42 \cdot 5 + 24 \\ 42 &= 24 \cdot 1 + 18 \\ 24 &= 18 \cdot 1 + 6 \\ 18 &= 6 \cdot 3 + 0 \end{aligned}$$

Therefore, $\gcd(234, 42) = 6$.

The counterpart to the greatest common divisor is the *least common multiple*.

Definition (Least Common Multiple). Let $a, b \in \mathbb{Z}^+$. $l \in \mathbb{Z}^+$ is the least common multiple of a and b, denoted $\operatorname{lcm}(a,b)$ if

1. $a \mid l$ and $b \mid l$
2. If $a \mid m$ and $b \mid m$ for some $m \in \mathbb{Z}^+$, then $l \leq m$.

One nice property of lcm and gcd is

$$\gcd(a,b) \cdot \operatorname{lcm}(a,b) = ab.$$

A interesting preliminary we should know before continuing onto groups is the *Fundamental Theorem of Arithmetic*.

Theorem (Fundamental Theorem of Arithmetic). Every integer $n > 1$ is either prime or is the product of prime numbers.

This can be proved using mathematical induction.

0.2 Equivalence Relations

Let us first introduce the concept of *ordered pairs*. When we say ordered pair in this context, we mean that if a and b are sets, then the ordered pair $(a,b) = \{\{a\},\{a,b\}\}$.

Exercise

Prove that $(a,b) = (c,d)$ *if* $a = c$ *and* $b = d$.

Let's gain a better understanding of this concept through an example.

Example 4. Consider the ordered pair $(0,1)$. In terms of set construction, $0 = \emptyset$ and $1 = \{\emptyset\}$. Therefore, the ordered pair $(0,1)$ is

$$(0,1) = \{\{0\},\{0,1\}\} = \{\{\emptyset\},\{\emptyset,\{\emptyset\}\}\}.$$

On the other hand, the ordered pair $(0,0)$ is defined as

$$(0,0) = \{\{\emptyset\},\{\emptyset,\emptyset\}\} = \{\emptyset\}.$$

Next we will define the *Cartesian product* of two sets.

Definition (Cartesian Product). Let A and B be sets. Then, the Cartesian product of A and B is

$$A \times B = \{(a,b) \mid a \in A \text{ and } b \in B\}.$$

The Cartesian product is simply all combinations of ordered pairs possible in two sets. With the notion of the Cartesian product, we can define relations

Definition (Relation). A subset of $A \times B$ is called a relation from A to B, i.e., R is a relation from A to B if and only if $R \subseteq A \times B$. If R is a relation from A to A itself, then R is called a relation on A.

Now we can define equivalence relations.

Definition (Equivalence Relation). Let R be a relation on a set A. Then R is called an equivalence relation if

1. $(a,a) \in R$ for all $a \in A$ (reflexive property)
2. $(a,b) \in R$ implies $(b,a) \in R$ for all $a,b \in A$ (symmetric property)
3. $(a,b) \in R$ and $(b,c) \in R$ implies $(a,c) \in R$ for all $a,b,c \in A$ (transitive property)

Do note the following.

Remark 1. If R is an equivalence relation, we write $a \sim b$ or $a \sim_R b$ as $(a,b) \in R$. It is also possible to write $a \approx b$ or $a \equiv b$.

Let's illustrate the concept of equivalence relations with an example.

Example 5. Let $A = \mathbb{Z}$ and $R_2 = \{(a,b) \in A \times A : 2 \mid a-b\}$. Let's verify the properties of an equivalence relation to see if R_2 is indeed one.

1. *Reflexivity.* Let $a \in \mathbb{Z}$. Then $a - a = 0$. We have $2 \mid 0$. Thus, $(a,a) \in R_2$.
2. *Symmetric.* Let $a, b \in \mathbb{Z}$ and $(a,b) \in R_2$. We have $2 \mid a-b$. That means there is $k \in \mathbb{Z}$ such that $a - b = 2k$. Therefore, $b - a = 2(-k)$ and $-k \in \mathbb{Z}$. We also have that $2 \mid b-a$. Thus $(b,a) \in R_2$.
3. *Transitivity.* Assume that $(a,b),(b,c) \in R_2$. Thus, $2 \mid a-b$ and $2 \mid b-c$. Therefore, there are $k_2, k_3 \in \mathbb{Z}$ such that (1) $a - b = 2k_2$ and (2) $b - c = 2k_3$. Adding (1) and (2), we have $a - c = 2(k_2 + k_3)$ and $k_2 + k_3 \in \mathbb{Z}$. Thus, $(a,c) \in R_2$.

Therefore, R_2 is an equivalence relation.

As a better example, *equality* (=) is an equivalence relation that works on *any* set. It is essentially the prototype of an equivalence relation. On the other hand, $(\mathbb{Z}, <)$ is not an equivalence relation, as $1 < 2$ but $2 \not< 1$, so it fails to satisfy reflexivity[‡].

Definition (Modulus Congruence). Let $n \in \mathbb{Z}^+$. Let $a, b \in \mathbb{Z}$. If $n \mid a-b$, then we write $a \equiv b (\bmod n)$ to mean "a is *congruent* to b modulo n".

Next we will define another useful tool related to relations.

Definition (Partition). A *partition* on a set S is a collection of nonempty disjoint subsets of S whose union is S.

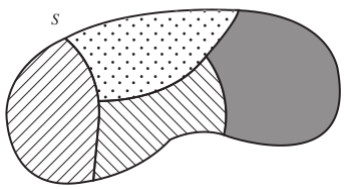


Figure 3: A partition of S into four subsets

As a quick notational remark, let Λ denote the index set and $P = \{A_i \mid A_i \subseteq S \text{ for } i \in \Lambda\}$, we say that P is a partition of S if

1. $A_i \neq \emptyset$ for all $i \in \Lambda$
2. $A_i \cap A_j = \emptyset$ for $i \neq j$
3. $\bigcup_{i \in \Lambda} A_i = S$

[‡]It also fails to satisfy the symmetric property, despite satisfying transitivity

Example 6. Let $S = \mathbb{Z}$ and $P = \{\mathbb{Z}^-, \mathbb{Z}^+, \{0\}\}$. This is a partition of the integers into negatives, positives and zero.

Example 7. If $S = \mathbb{Z}$, then $\{S\}$ is a partition of S (perhaps the *laziest* partition we could construct).

Example 8. Suppose that $n = 3$ and $S = \mathbb{Z}$, and we define each A_i as

$$\begin{aligned} A_0 &= \{0, \pm 3, \pm 6, \pm 9\} \\ A_1 &= \{\dots, -5, 2, 1, 4, 7, 10 \dots\} \\ A_2 &= \{\dots, -4, -1, 2, 5, 8, 11, \dots\} \end{aligned}$$

then we can define $P = \{A_0, A_1, A_2\}$ using modulo 3.

The next theorem reveals that equivalence relations and partitions are deeply intertwined.

Theorem 0.7 (Equivalence Classes Partition). The equivalence classes of an equivalence relation on a set S constitute a partition of S. Conversely, for any partition P of S, there is an equivalence relation on S whose equivalence classes are the elements of P.

Proof. Let $\sim_R$ be an equivalence relation on a set S. We define a collection of subsets of S as follows

$$P = \{[a] \mid a \in S\}.$$

We will show that P is a partition.

1. $\emptyset \notin P$ – let $a \in S$. Since $a \sim a$, we have $a \in [a]$, i.e., $[a] \neq 0$. Thus, $\emptyset \notin P$.
2. $A_i \cap A_j$ for $i \neq j$ – assume that $[a]$ and $[b]$ are equivalence classes such that $[a] \neq [b]$. We want to show that $[a] \cap [b] \neq \emptyset$. We want to show that $[a] \cap [b] \neq \emptyset$. Assume to the contrary that $[a] \cap [b] \neq \emptyset$. Let $c \in [a] \cap [b]$. If $a_1 \in [a]$ and $b_1 \in [b]$, then $a_1 \sim c$ and $c \sim b_1$. Therefore, $a_1 \sim b_1$ by transitivity of $\sim$. Thus, $[a] = [b]$.
3. $\bigcap_{a \in S} [a] = S$.

To prove the converse, let P be a partition of a set S, say, $P = \{A_i \mid A_i \subseteq S, i \in \Lambda\}$. Define a relation R on S as follows – for any $a, b \in S$, $(a, b) \in R$ if there is $i \in \Lambda$ such that $(a, b) \in A_i$. We can show that this is an equivalence relation. □

Chapter 1

Introduction to Groups

A Brief History of Abstract Algebra

In the time of the Babylonians (approximately 2000BC), people knew how to solve linear quadratic equations such as $ax + b = 0$ or $ax^2 + bx + c = 0$. Note that naturally, they did not have the same notation we do today, *nor is it likely they had paper to write on*.

Cubic equations were only later tackled by mathematicians like Cardano, Fontana, and Ferro around 1500BC. Further on, Ferrari (most likely not the same guy who made the supercar) discovered how to solve quartic equations. The next challenge became the *quintics*. Mathematicians struggled to solve the solution for quintics for hundreds of years.

It was later discovered by Abel and Galois that quintics have *no solutions*. They did so using *groups*. It became quickly realized that the abstract concept of groups was widely applicable to various fields of mathematics.

1.1 Symmetries of a Square

Consider the following square

4	1
3	2

In its current state, if we were to overlay a copy of itself behind it, we'd have the mapping

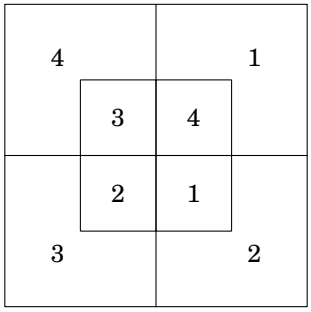
$$\begin{array}{cccc} 1 & 2 & 3 & 4 \\ \updownarrow & \updownarrow & \updownarrow & \updownarrow \\ 1 & 2 & 3 & 4 \end{array}$$

This is akin to an *identity function*[*], which we'll denote

$$e = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 1 & 2 & 3 & 4 \end{pmatrix}.$$

[*]Note that it is both surjective and injective.

Suppose we were to rotate this square clockwise by 90°.



We would then have the mapping

$$R_{90} = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix}.$$

Performing four rotations, we'd have

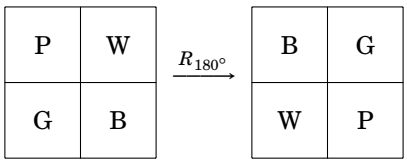
$$(R_{90})^2 = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix}$$

$$(R_{90})^3 = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix}$$

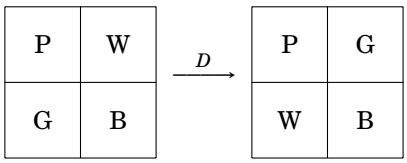
$$(R_{90})^4 = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 1 & 2 & 3 & 4 \end{pmatrix}$$

Note that $\left(R'_{90}\right)^4 = e$. We can perform the following actions on this square

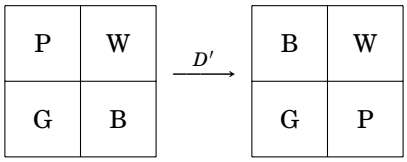
- Rotation in increments of 90° (denoted R)



- Spinning the square along the main diagonal (denoted D)



- Spinning the square about the other diagonal denoted (D')



- Flipping the square about the horizontal axis (denoted H)

P	W
G	B

$\xrightarrow{H}$

G	B
P	W

- Flipping the square about the vertical axis (denoted V)

P	W
G	B

$\xrightarrow{V}$

W	P
B	G

The study of group theory is all about exploring what happens when we perform chains of these operations and the resulting effects on our shape of interest. For example, consider

P	W
G	B

$\xrightarrow{R_{90}}$

W	B
P	G

$\xrightarrow{H}$

P	G
W	B

This pair of motions, taken together, is equal to the single motion D. This suggests we can compose two motions to obtain a single motion, and indeed we can. We can construct an *operation table*, known as a *Cayley table*[†].

	R_0	R_{90}	R_{180}	R_{270}	H	V	D	D'
R_0	R_0	R_{90}	R_{180}	R_{270}	H	V	D	D'
R_{90}	R_{90}	R_{180}	R_{270}	R_0	D'	D	H	V
R_{180}	R_{180}	R_{270}	R_0	R_{90}	V	H	D'	D
R_{270}	R_{270}	R_0	R_{90}	R_{180}	D	D'	V	H
H	H	D	V	D'	R_0	R_{180}	R_{90}	R_{270}
V	V	D'	H	D	R_{180}	R_0	R_{270}	R_{90}
D	D	V	D'	H	R_{270}	R_{90}	R_0	R_{180}
D'	D'	H	D	V	R_{90}	R_{270}	R_{180}	R_0

Table 1.1: The Cayley table for our square.

We notice a couple of things from this table. First, any sequence of motions results in one of the eight possible motions. We call this property *closure*. Next, notice that R_0 combined with any other motion is the motion itself, regardless of the order R_0 is applied. An element such as R_0 is called an *identity*. Moreover, for each element A in the possible eight, there exists an element B such that $AB = BA = R_0$. In case case, B is said to be the *inverse* of A (and vice versa). For example, R_{90} and R_{270} are each others inverse.

[†]Named after the English mathematician Arthur Cayley, who first introduced them in 1854.

1.2 The Dihedral Groups

The analysis we just performed for a square can be similarly done for an equilateral triangle or regular pentagon. In fact, we can do this for any regular n-gon, where $n \geq 3$. The corresponding group is denoted D_n and called *the dihedral group of order* $2n$. We do not have to restrict ourselves to two dimensions, as well. The rotation of a plane about a point in the plane is a symmetry of the plane. Reflections/translations can be defined similarly.

Dihedral groups arise frequently in art and nature. Chemists classify molecules by their symmetry[‡]. Symmetry considerations are even applied in orbital calculations, determining energy levels of atoms and molecules, and in the study of molecular vibrations.

Figure 1.1: Some members of *Echinodermata* phylum (starfish, sea urchins, and sand dollars). Most members display D_5 symmetry, however, the Sun Starfish on the bottom right has D_{13} symmetry. [Source]

Dihedral groups are an example of a *group*. In the next chapter, we introduce a formal definition for what groups are and some properties they have.

[‡]The term *symmetry* comes from the Greek *symmetros*, meaning "of like measure".

Chapter 2

Groups

2.1 Definition and Examples

The term *group* was introduced by Évariste Galois around 1830 to describe sets of one-to-one functions on finite sets that could be grouped together to form a set closed under composition. Before getting to their definition, let us introduce the concept of a *binary operation*.

Definition (Binary Operation). A *binary operation* on a set G is a function $* : G \times G \to G$ that assigns each ordered pair of elements of G an element of G.

As a quick remark, note that $f : G \to G$ is called a *unary operation*[*] on G. Most common operations in mathematics are binary, e.g., for $\mathbb{Z}^+$, addition and multiplication are binary operations. Observe that we are restricting ourselves to the set of positive integers as we want to maintain the property of *closure*. Operations like subtraction and division are *not* binary operations on $\mathbb{Z}^+$, as they do not satisfy closure. For example

$$1 - 2 = -1 \notin \mathbb{Z}^+.$$

Now, we can introduce the core idea of the class: *groups*!

Definition (Group). Let G be a set together with a binary operation (usually called multiplication) that assigns to each ordered pair (a, b) of elements of G an element in G denoted by ab. We say that G is a *group* under this operation if the following three properties are satisfied.

i) *Associativity*. The operation is associative; that is, $(ab)c = a(bc)$ for all a, b, c in G.

ii) *Identity*. There is an element e (called the *identity*) in G such that $ae = ea = a$ for all a in G.

iii) *Inverses*. For each element a, there is an element b in G (called an *inverse* of a) such that $ab = ba = e$.

We say that if $ab = ba$ for all $a, b \in G$, then G is called an *Abelian group*[†].

[*] From the Latin *unus* meaning one.

[†] Named after the 19th century mathematician Niels Henrik Abel.

Example 9. $(\mathbb{Z},+),(\mathbb{Q},+),(\mathbb{R},+),(\mathbb{C},+)$ are all examples of abelian groups. On the other hand, $(\mathbb{Z},\cdot)$ is *not* a group. Despite associativity being satisfied and the existence of the identity element 1, for any $b \in \mathbb{Z}$, we can't satisfy

$$0 \times b = 1 = b \times 0.$$

In other terms, $0 \in \mathbb{Z}$ does not have a multiplicative inverse.

2.2 Elementary Properties of Groups

Let us deduce some properties that all groups share, beginning with the fact that the identity element is unique.

Theorem 2.1 (Uniqueness of Identity). Let $(G,*)$ be a group. G has a unique identity.

Proof. Assume there are two identity elements $e, e' \in G$. We have that

$$\underbrace{e * e' = e'}_{e \text{ is an identity}} = \underbrace{e' * e = e}_{e' \text{ is an identity}}$$

Thus, we have that $e' = e$. □

Due to this theorem, we may unambiguously refer to "the identity" of a group. It is commonly denoted e as the German word for identity is *Einheit*[‡].

Theorem 2.2 (Cancellation). In a group G, the right and left cancellation laws hold; that is, $ba = ca$ implies $b = c$ and $ab = ac$ implies $b = c$.

Proof. Assume that $ab = ac$. Thus, we have that

$$\begin{aligned} a^{-1}(ab) &= a^{-1}(ac) \\ (a^{-1}a)b &= (a^{-1}a)c \\ eb &= ec \\ b &= c. \end{aligned}$$

□

One consequence of the cancellation property is the fact that each group element occurs exactly once in each row and column of a group's Cayley table. Another is the *uniqueness of inverses*.

Theorem 2.3 (Uniqueness of Inverses). For each element a in a group G, there is a unique element b in G such that $ab = ba = e$.

[‡]That's awesome as hell.

Proof. We can pretty much prove existence for free. Let $a \in G$. Since G is a group, a has an inverse, say b. Let $b' \in G$ be another inverse of G. Since b and b' are both inverses of a, we have that

$$ab = e = ab \tag{1}$$

Consider $b \times (1)$

$$b(ab) = be = b(ab').$$

We can then conclude that

$$\begin{aligned} b(ab) &= b(ab') \\ (ba)b &= (ba)b' \\ eb &= eb\prime \\ b &= b'. \end{aligned}$$

□

Note that for $a \in G$, we will denote the inverse of a as a^{-1}. This notation is similar to that used for ordinary real numbers under multiplication. We n is a positive integer, the associative property of groups allows us to use g^n to denote

$$\underbrace{gg\cdots g.}_{n \text{ factors}}$$

We also define $g^0 = e$. For negative n, we define $g^n = (g^{-1})^{|n|}$. Note that in an abstract group, we don't have the notion of noninteger exponents. However, the general rules of exponentiation hold for groups. Note that when working with groups that use addition as the group operation, we would have that

$$g^n = \underbrace{g + g + \cdots + g.}_{n\text{terms}}$$

We usually denote this as gn instead (although it should not be interpreted as combining n and g under the group operation, as n may not even be an element of the group). Finally, although groups do not have the property that $(ab)^n = a^n b^n$, there is a simple relationship between $(ab)^{-1}$ and a^{-1} and b^{-1}.

Theorem 2.4 (Socks-Shoes Property). For group elements a and b, $(ab)^{-1} = b^{-1}a^{-1}$.

Proof. Let G be a group and let $a, b \in G$. Consider the equation

$$(ab)(ab)^{-1} = e. \tag{1}$$

We have also that

$$\begin{aligned} (ab)(b^{-1}a^{-1}) &= a(bb^{-1})a^{-1} \\ &= aa^{-1} \\ &= e. \end{aligned} \tag{2}$$

From (1) and (2), we have that

$$(ab)(ab)^{-1} = (ab)(b^{-1}a^{-1}),$$

and from the cancellation property

$$ab^{-1} = b^{-1}a^{-1}.$$

□

2.3 $\mathbb{Z}_n$ and $U(n)$

Two groups we will be studying in great detail throughout this course are $\mathbb{Z}_n$ and $U(n)$. We define $\mathbb{Z}_n$ as

$$\mathbb{Z}_n = \{\overline{0}, \overline{1}, \ldots, \overline{n-1}\},$$

where $\overline{k} = \{r \in \mathbb{Z} : n \mid k - r\}$ or $r \equiv k \pmod{n}$. In other terms, the set of integers from 0 to $n-1$ under addition modulo n. Somewhat similarly, we define $U(n)$ as

$$U(n) = \{\overline{i} \in \mathbb{Z}_n \mid \exists \overline{j} \in \mathbb{Z}, \overline{i} \cdot \overline{j} = \overline{1}\}.$$

In other terms, it is the set of integers from 1 to $n-1$ that are relatively prime to n. Let's prove that $U(n)$ is a group. We have to show that

1. $U(n)$ is closed under multiplication. Let $a, b \in U(n)$. Thus, there are $\overline{a'}, \overline{b'} \in \mathbb{Z}_n$ such that $\overline{a} \cdot \overline{a'} = \overline{1}$ and $\overline{b} \cdot \overline{b'} = \overline{1}$. We want to show that $\overline{a} \cdot \overline{b} = \overline{ab} \in U(n)$.

$$\begin{aligned}
\overline{ab} \cdot \overline{a'b'} &= \overline{(ab) \cdot (a'b')} \\
&= \overline{(ab') \cdot (bb'} \\
&= \overline{1 \cdot 1} \\
&= \overline{1} \in U(n)
\end{aligned}$$

2. That $U(n)$ satisfies associativity.

$$\begin{aligned}
(\overline{a} \cdot \overline{b}) \cdot \overline{c} &= (\overline{a \cdot b}) \cdot \overline{c} \\
&= \overline{a \cdot b \cdot c} \\
&= \overline{a} \cdot (\overline{b \cdot c})
\end{aligned}$$

3. $U(n)$ has an inverse. Let $\overline{a} \in U(n)$. There is $\overline{b} \in \mathbb{Z}_n$ such that $\overline{a} \cdot \overline{b} = \overline{1}$. Consider $\overline{b} \in \mathbb{Z}_n$. There is $\overline{a} \in \mathbb{Z}_n$ such that $\overline{a} \cdot \overline{b} = \overline{1} = \overline{b} \cdot \overline{a}$. Thus, $\overline{b} \in U(n)$.

2.4 Bézout's Lemma

One final result we will introduce is *Bézout's Lemma*, which serves as a useful tool for doing proofs down the line.

Theorem (Bézout Lemma). Let $a, b \in \mathbb{Z}$ such that one of them is nonzero. If $\gcd(a, b) = 1$, then there are integers $x, y \in Z$ such that $ax + by = 1$.

More generally, we have that for $a, b \in \mathbb{Z}$, there exist $x, y \in \mathbb{Z}$ such that $ax + by = \gcd(a, b)$.

Chapter 3

Finite Groups and Subgroups

3.1 Terminology and Notation

Finite groups, i.e., groups with finitely many elements, have interesting arithmetic properties. Before studying them, we will introduce some new terminology and notation.

Definition (Order of a Group). The number of elements of a group (finite or infinite) is called its *order*. We will use $|G|$ to denote the order of G.

The group $(\mathbb{Z},+)$ has infinite order, whereas the group $U(10) = \{1,3,7,9\}$ under multiplication modulo 10 has order 4.

Definition (Order of an Element). The *order* of an element g in a group G is the smallest positive integer n such that $g^n = e$. (In additive notation, this would be $ng = 0$.) If no such integer exists, we say that g has infinite order. The order of an element g is denoted by $|g|$.

The order of a group element g can be found by computing $g, g^2, g^3, \ldots$ until the identity is reached. If the identity never appears, we say that g has infinite order. We also note that some groups are subsets of others with the same operation. These are called *subgroups*.

Definition (Subgroup). If a subset H of a group G is itself a group under the operation of G, we say that H is a *subgroup* of G.

We use the notation $H \leq G$ to denote that H is a subgroup of G. If we want to indicate that H is a subgroup of G but is not equal to G itself, we write $H < G$. Such a subgroup is called a *proper subgroup*. The subgroup $\{e\}$ is called the *trivial subgroup* of G; a subgroup that is not $\{e\}$ is called a *nontrivial subgroup* of G.

Example 10. Consider the group

$$G = \{A \in M_{2\times 2}(\mathbb{R}) \mid \det(A) \neq 0\}.$$

Now consider the following set

$$H = \{A \in M_{2\times 2}(\mathbb{Z}) \mid \det(A) \neq 0\}.$$

Is this H a subgroup of G? The answer is *no*. Consider the matrix

$$A = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}.$$

The determinant of this matrix is

$$\det(A) = 2 \cdot 1 - 0 \cdot 0 = 2.$$

The inverse of A is

$$A^{-1} = \begin{pmatrix} 1 & 0 \\ 0 & \frac{1}{2} \end{pmatrix}.$$

We can confirm this

$$\begin{aligned} AA^{-1} &= \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & \frac{1}{2} \end{pmatrix} \\ &= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \\ &= I. \end{aligned}$$

However $A^{-1} \notin M_{2\times 2}(\mathbb{Z})$.

We've seen that although the set $H = \{A \in M_{2\times 2}(\mathbb{Z}) \mid \det(A) \neq 0\}$ was indeed a subset of G, it was not a subgroup. What about the set

$$H_2 = \{A \in M_{2\times 2}(\mathbb{Z}) \mid \det(A) = 1\} \subseteq G.$$

Let's see if it satisfies the condition for a subgroup.

1. First let's show closure is satisfied. We have that for $A, B \in H_2$

$$\begin{aligned} \det(AB) &= \det(A)\det(B) \\ &= 1 \cdot 1 \\ &= 1 \in H_2. \end{aligned}$$

2. For associativity, we have that since $H_2 \subseteq G$, it has associativity under the same operation "$\cdot$".

3. For the identity, we have that

$$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

is an identity with $\det(I) = 1$, i.e., $I \in H_2$.

4. For the inverse, consider the matrix

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in H_2.$$

Thus, $a,b,c,d \in \mathbb{Z}$ with $\det(A) = ad - bc$.

$$\begin{aligned} A^{-1} &= \frac{1}{ad-bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix} \\ &= \frac{1}{1}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix} \\ &= \begin{pmatrix} d & -b \\ -c & a \end{pmatrix} \in M_{2\times 2}(\mathbb{Z}). \end{aligned}$$

Observe that

$$\det\left(A^{-1}\right) = \frac{1}{\det(A)} = \frac{1}{1} = 1.$$

Thus, $AA^{-1} = I_{2\times 2}A^{-1}A$ and $\det\left(A^{-1}\right) = 1$.

Therefore, we can conclude that $H_2 \leq G$. The set H_2 itself actually has a name: $SL_2(\mathbb{Z})$*.

3.2 Subgroup Tests

From the examples we've done so far, showing that a set is a subgroup is not always a trivial procedure. Luckily, we have some tests that we can do to make our lives easier.

Theorem 3.1 (One-Step Subgroup Test). Let G be a group and H a nonempty subset of G. If ab^{-1} is in H whenever a and b are in H, then H is a subgroup of G. (In additive notation, if $a - b$ is in H whenever a and b are in H, then H is a subgroup of G.)

Proof. We will prove that these conditions satisfy all four properties of a group.

1. *Associativity*: since $H \subseteq G$, then the operation is associative in H.
2. *Identity*: since $H \neq \emptyset$, there is $x \in H$. Let $a = x = b$. By our assumption that $ab^{-1} \in H$, we have that
$$ab^{-1} = xx^{-1} = e \in H.$$
Therefore, H contains an identity element.
3. *Inverse*: let $y \in H$. Choose $a = e \in H$ and $b = y \in H$. By our assumption, ab^{-1} must be in H as well, i.e.,
$$ab^{-1} = ey^{-1} = y^{-1} \in H.$$
Hence, y has the inverse $y^{-1} \in H$.
4. *Closure*: let $x_1, x_2 \in H$. We want to show that $x_1x_2 \in H$. We know from (3) that $x_2^{-1} \in H$. We choose $a = x_1$ and $b = x_2^{-1}$ so by the assumption above
$$ab^{-1} = x_1(x_2^{-1})^{-1} = x_1x_2 \in H.$$

□

Let us now apply this theorem with an example.

*Read as Special Linear Group over $\mathbb{Z}$.

Example 11. Let G be an abelian group. Define $H = \{x \in G \mid x^2 = e\}$. We will show that $H \leq G$. Note that H is non-empty as $e^2 = e$, i.e., $e \in H$. We can now apply the One-Step Subgroup Test. Assume that $a, b \in H$. That means $a^2 = e = b^2$. We want to show that ab^{-1} is also in H, i.e., $(ab^{-1})^2 = e$.

$$\begin{aligned}
(ab^{-1})^2 &= (ab^{-1})(ab^{-1}) \\
&= (aa)(b^{-1}b^{-1}) && \text{(abelian)} \\
&= a^2(b^2)^{-1} \\
&= ee^{-1} \\
&= ee \\
&= e.
\end{aligned}$$

Thus, $ab^{-1} = e$. We have shown that $ab^{-1} \in H$. By the one-step subgroup test, we have that H is a subgroup of G.

An easier variant of the One-Step Subgroup Test is, ironically, the *Two-Step Subgroup Test*.

Theorem 3.2 (Two-Step Subgroup Test). Let G be a group and let H be a nonempty subset of G. If ab is in H whenever a and b are in H (H is closed under the operation), and a^{-1} is in H whenever a is in H (H is closed under taking inverses), then H is a subgroup of G.

Proof. The proof follows from the One-Step Subgroup Test. □

Example 12. Let G be an abelian group. Define $H = \{a \in G \mid |a| \text{ is finite}\}$. Prove that $H \leq G$.

How do we show that $H \neq \emptyset$? Since $|e| = 1 < \infty$, $e \in H$. Thus, $H \neq 0$. Next, we show H is closed. Let $a, b \in H$. We have that $|a| = m \in \mathbb{N}$ and $|b| = n \in \mathbb{N}$. In other terms, $a^m = e$ and $b^n = e$. We have that

$$\begin{aligned}
(ab)^{mn} &= \underbrace{(ab)\ldots(ab)}_{mn \text{ times}} \\
&= a^{mn}b^{mn} \\
&= (a^m)^n(b^n)^m \\
&= e^n e^m \\
&= ee \\
&= e \in H.
\end{aligned}$$

This means that $|ab|$ is finite. To show that H contains an inverse, let $x \in H$, i.e., $|x| == k < \infty$, or $x^n = e$. Consider

$$(x^{-1})^k = (x^k)^{-1} = e^{-1} = e.$$

Thus, $|x^{-1}|$ is finite. Therefore, by the two-step subgroup test, $H \leq G$.

We will introduce a final test we can use – the finite subgroup test.

Theorem (Finite Subgroup Test). Let G be a group and $\emptyset \neq H \subseteq G$. If H is finite and $\forall a, b \in H$, $ab \in H$, then $H \leq G$.

Proof.

1. *Closure*: closure is satisfied by our assumption of H.
2. *Inverse*: let $a \in H$. We want to show that $a^{-1} \in H$. Consider $S = \{a^k \in H \mid k \in \mathbb{N}\}$. We have that $S \subseteq H$, since H is finite, S is finite as well, i.e.,

$$S = \{a, a^2, a^3, \dots\}.$$

Thus, there exists $a^m = a^n$ for some $m, n \in \mathbb{N}$, and $m < n$ (otherwise it would be infinite if they did not repeat). So, we have

$$\begin{aligned} \underbrace{a \dots a}_{m \in \mathbb{N}} &= \underbrace{a \dots a}_{m \in \mathbb{N}} \cdot \underbrace{a \dots a}_{n-m \in \mathbb{N}} \\ a^m &= a^m \cdot a^{n-m} \\ e &= a^{n-m}. \end{aligned}$$

If $n - m = 1$, then $e = a$, we are done. If $n - m > 1$, then $e = a^{n-m} = a \cdot a^{n-m-1}$. Choose $b = a^{n-m-1} \in S \subseteq H$. We have $b = a^{-1}$, as $ab = e = ba$. Both cases imply that $a^{-1} \in H$.

□

How do we show that a subset of a group is *not* a subgroup? We can

1. Show that the identity is not in the set.
2. Show there exists an element of the set whose inverse is not in the set.
3. Show there exists two elements of the set whose product is not in the set.

3.3 Examples of Subgroups

Let us now introduce an important notation. For any element a in a group, we define $\langle a \rangle$ to be the set $\{a^n \mid n \in \mathbb{Z}\}$. Observe that the exponents of a include all negative integers as well as 0 and the positive integers. Next we will introduce a new definition for the center of a group.

Theorem 3.4 ($\langle a \rangle$ Is a Subgroup). Let G be a group, and let a be any element of G. Then, $\langle a \rangle$ is a subgroup of G.

Proof. Since $a \in \langle a \rangle$, it is non-empty. Let $a^n, a^m \in \langle a \rangle$. Then $a^n (a^m)^{-1} \in \langle a \rangle$, so by the One-Step Subgroup Test, $\langle a \rangle$ is a subgroup of G. □

We call the subgroup $\langle a \rangle$ the *cyclic subgroup of G generated by a*. In the case that $G = \langle a \rangle$, we say that G is *cyclic* and a is a *generator* of G[†]. We note that every cyclic group is Abelian[‡]. Next, we consider one of the most important subgroups.

[†]Note that a cyclic group may have many generators.
[‡]Why? Since $a^i a^j = a^{i+j} = a^{j+i} = a^j a^i$.

Definition (Center of a Group). Let G be a group. Define $Z(G) = \{a \in G \mid \forall x \in G, ax = xa\}$. Note that G is not necessarily abelian. $Z(G)$ is called the *center* of G.

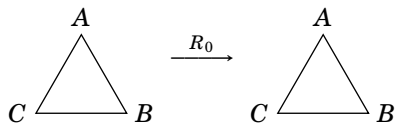
In other terms, $Z(G)$[§] is the subset of elements in G that commute with every element in G. Let's illustrate this concept with some examples.

Example 13. G is an abelian group. Then for every $a \in G$ and $x \in G$, $ax = xa$. Therefore, $Z(G) = G$.

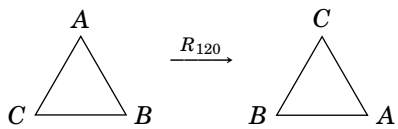
What about a non-abelian group?

Example 14. Consider the dihedral group D_3. D_3 has six operations, which are the following

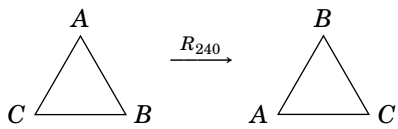
1. The identity operation R_0



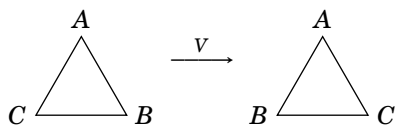
2. Rotation by 120 degrees, denoted R_{120}



3. Rotation by 240 degrees, denoted R_{240}

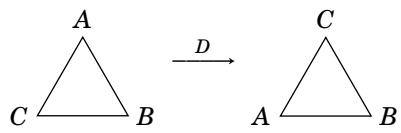


4. Flipping about the vertical axis

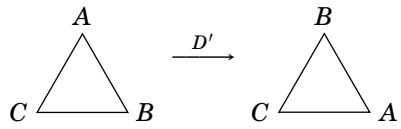


5. Flipping about the main diagonal

[§] Z comes from the fact that the German word for center is *zentrum*.



6. Flipping about the other diagonal



We have that $(V)^2 = (D)^2 = (D')^2 = e$, so these operations have order 2. As for the rotations: $|R_0| = 1$, since $R_0 = e$; $|R_{120}| = 3$, since $(R_{120})^3$; and $|R_{240}| = 3$, since $(R_{240})^3 = e$. Therefore, D_3 has the set of operations $\{V, D, D', R_0, R_{120}, R_{240}\}$, $|D_3| = 6$.

As $R_1V_1 \neq V_1R_1$, G is not abelian. Note that for any rotation R and reflection V, $RV \neq VR$. Therefore, $Z(G) = \{e\}$. Why is this the case? Let $x \in G \setminus \{e\}$. Then x is a rotation or reflection. Choose y to be the other type of symmetry (i.e., if x is rotation, y is reflection, and vice versa).

Next we show that the center of a group is a subgroup.

Theorem 3.5 (Center Is a Subgroup). The center of a group G is a subgroup of G.

Proof. We can use the two-step subgroup test.

1. *Identity*: let $x \in G$. We have $ex = xe$. Thus, by definition, $e \in Z(G)$.
2. *Invertibility*: let $a \in Z(G)$. We want to show that $a^{-1} \in Z(G)$ as well. For any $x \in G$, we have that $ax = xa$. We have that

$$\begin{aligned} ax &= xa \\ a^{-1}(ax) &= a^{-1}(xa) \\ a^{-1}(ax)a^{-1} &= a^{-1}(xa)a^{-1} \\ (a^{-1}a)xa^{-1} &= a^{-1}x(aa^{-1}) \\ xa^{-1} &= a^{-1}x. \end{aligned}$$

Thus, $a^{-1} \in Z(G)$. By the two step subgroup test, $Z(G) \leq G$.

□

Although an element from a non-Abelian group does not necessarily commute with every element of the group, there are usually some elements with which it does. This leads us to the next definition and theorem.

Definition (Centralizer of a in G). The *centralizer* of $a \in G$ is defined by $C(a) = \{g \in G \mid ag = ga\}$.

In other terms, $C(a)$ is the set of all elements in G that commute with a. Let's follow this up with an example.

Example 15. If $G = D_3$ and $a = R_1$, then

$$\begin{aligned} C(a) &= C(R_1) \\ &= \{e, R_1, R_1^2\} \\ &= \langle R_1 \rangle \end{aligned}$$

If $a = V_2$, then

$$C(V_2) = \{e, V_2\}$$

Similarly to $Z(G)$, $C(a)$ is also a subgroup.

Theorem 3.6 (C(a) Is a Subgroup). For $a \in G$, $C(a) \leq G$.

Proof. Similar to the previous theorem. □

Chapter 4

Cyclic Groups

In this chapter, we will examine cyclic groups in detail and determine their important characteristics. Recall the definition of a cyclic group from the previous chapter.

Definition (Cyclic Group). G is a *cyclic group* if there is $a \in G$ such that $G = \{a^n \mid n \in \mathbb{Z}\} = \langle a \rangle$. In this instance, a is called a *generator* of G.

We can begin with a few examples.

Example 16. Consider the group $(\mathbb{Z}, +)$, with identity zero. Let $n \in \mathbb{Z}$. For $n \geq 1$, we have

$$n = \underbrace{1 + 1 + 1 + \cdots + 1}_{n \text{ times}}.$$

For $n = 0$, we have

$$1^0 = e = 0.$$

For $n \leq -1$, i.e., $n = -k$ for some $k \in \mathbb{N}$, we have

$$\begin{aligned} n &= -k \\ &= 1^{-k} \\ &= (1^{-1})^k \\ &= (-1)^k \\ &= \underbrace{(-1) + \cdots + (-1)}_{k \text{ times}} \end{aligned}$$

Thus, we may conclude that $(\mathbb{Z}, +) = \langle 1 \rangle$. Similarly, we can show that $(\mathbb{Z}, +) = \langle -1 \rangle$.

We've noticed two things from the previous example: the inverse of a generator is also a generator, and group notation is pretty complicated. This is worth remarking.

Remark 2.

1. If $G = \langle a \rangle$, then $G = \langle a^{-1} \rangle$, as $a = (a^{-1})^{-1}$.

2. If G is abelian, we can use an "additive" version of x^n, i.e.,

$$x^n = x + x + \cdots + x = nx.$$

Let's consider another example.

Example 17. Consider $(\mathbb{Z}_n, +) = \langle \overline{1} \rangle = \langle \overline{-1} \rangle$. Recall that

$$\mathbb{Z}_n = \{\overline{0}, \overline{1}, \overline{2}, \ldots, \overline{n-1}\}$$

What about a specific n? Consider $n = 6$.

$$\mathbb{Z}_6 = \{\overline{0}, \overline{1}, \ldots, \overline{5}\} = \langle \overline{1} \rangle = \langle \overline{-1} \rangle \equiv \langle \overline{5} \rangle.$$

Are there any other generators?

$$\begin{aligned}
\langle \overline{0} \rangle &= \{\overline{0}\rangle \\
\langle \overline{2} \rangle &= \{\overline{0}, \overline{2}, \overline{4}\} \\
\langle \overline{3} \rangle &= \{\overline{0}, \overline{3}\} \\
\langle \overline{4} \rangle &= \{\overline{0}, \overline{4}, \overline{2}\} \\
\langle \overline{5} \rangle &= \{\overline{0}, \overline{5}, \overline{10}, \overline{9}, \overline{8}, \overline{7}\} \\
&= \{\overline{0}, \overline{5}, \overline{4}, \overline{3}, \overline{2}, \overline{1}\} \\
&= \mathbb{Z}_6
\end{aligned}$$

Therefore, $\mathbb{Z}_6 = \langle \overline{1} \rangle, \langle \overline{5} \rangle$.

To generalize the previous example somewhat, in $\mathbb{Z}_n$, we have that $\langle \overline{a} \rangle = \{k\overline{a} \mid k \in \mathbb{Z}\}$. We see that $\langle \overline{a} \rangle = \mathbb{Z}_n$ if for any $\overline{x} \in \mathbb{Z}_n$ there is $k \in \mathbb{Z}$ such that $\overline{x} = k\overline{a}$. In other words, $x - ka$ is divisible by n, i.e., $n \mid x - ka$.

Example 18. Consider the set $\mathbb{Z}_{14} = \{\overline{0}, \overline{1}, \ldots, \overline{13}\}$. Let's consider the sets generated by each element.

$$\begin{aligned}
\langle \overline{0} \rangle &= \left\{k \cdot \overline{0} \mid k \in \mathbb{Z}\right\} \\
&= \left\{\overline{0}\right\} \\
\langle \overline{1} \rangle &= \mathbb{Z}_{14} \\
\langle 2 \rangle &= \left\{k \cdot \overline{2} \mid k \in \mathbb{Z}\right\} \\
&= \left\{\ldots, \overline{-2}, \overline{0}, \overline{2}, \overline{4}, \overline{6}, \ldots\right\} \\
&= \left\{\overline{0}, \overline{2}, \overline{4}, \ldots, \overline{10}, \overline{12}\right\} \\
\langle \overline{3} \rangle &= \left\{\overline{0}, \overline{3}, \overline{6}, \overline{9}, \overline{12}, \overline{15} \equiv 1, \overline{4}, \overline{7}, \overline{10}, \ldots\right\} \\
&= \mathbb{Z}_{14}.
\end{aligned}$$

So what exactly are the $\overline{s}$ such that $\langle \overline{s} \rangle = \mathbb{Z}_{14}$? Essentially, we want to find $\overline{s}$ such that

$$\langle \overline{s} \rangle = \left\{k \cdot \overline{s} \mid k \in \mathbb{Z}\right\} = \mathbb{Z}_{14}.$$

We can narrow our target down to finding $k \in \mathbb{Z}$ such that $k \cdot \overline{s} = \overline{1}$, i.e., $14 \mid k \cdot \overline{s} - 1$. This is because if we have the element $\overline{1}$, we can generate the entire set $\mathbb{Z}_{14}$. Equivalently, this

means there is $k\overline{s} - 1 = 14l$, for some $l \in \mathbb{Z}$. Rearranging, we have

$$\begin{aligned} k\overline{s} - 1 &= 14l \\ 1 &= 14(-l) + \overline{s}(-k) \\ 1 &= 14l_1 + \overline{s}k_1, \end{aligned}$$

where $l_1 = -l$ and $k_1 = -k$. Recall Bezout's Theorem, which states that if $a, b \in \mathbb{Z}$ such that one of them is nonzero and $\gcd(a,b) = 1$, then there are integers $x, y \in Z$ such that $ax + by = 1$. Therefore, we have that

$$1 = 14l_1 + \overline{s}k_1,$$

for some $l_1, k_1 \in \mathbb{Z}$ if and only if $\gcd(14, \overline{s}) = 1$. In conclusion, $\mathbb{Z}_{14} = \langle \overline{s} \rangle$ if and only if $\gcd(14, \overline{s}) = 1$.

What we just did in the example above is true in general.

Remark 3. If we work with any n, $\mathbb{Z}_n = \langle \overline{s} \rangle$ if and only if $\gcd(n, \overline{s}) = 1$.

Let's see if the same is true for $U(n)$.

Example 19. Consider the set

$$U(n) = \left\{ k \in \mathbb{Z}_n \mid \exists s \in \mathbb{Z}_n, \overline{k}\overline{s} = \overline{1} \right\}.$$

Consider $n = 10$. We have that

$$U(10) = \left\{ \overline{1}, \overline{3}, \overline{7}, \overline{9} \right\}.$$

This is because the greatest common denominator of every element in the set and 10 is equal to one. Another conclusion we can make this is that $\overline{s}$ in $U(10)$ has a multiplicative inverse if and only if $\gcd(s, 10) = 1$. Consider

$$\overline{3} \cdot \overline{-3} = \overline{3} \cdot \overline{7} = \overline{21} = \overline{1}.$$

Again, we can also generalize the example above.

Remark 4. $\overline{s}$ in $U(n)$ has an inverse if and only if $\gcd(s, n) = 1$. We can thus redefine $U(n)$ as follows

$$U(n) = \left\{ \overline{k} \in \mathbb{Z}_n \mid \gcd(k, n) = 1 \right\}.$$

This definition is a bit less computationally expensive, as we only need to compute the greatest common denominator instead of finding inverses. Let's show an example of a *non-cyclic* group.

Example 20. $U(8)$ contains the elements

$$\left\{ \overline{1}, \overline{3}, \overline{5}, \overline{7} \right\}.$$

Trivially, $\langle \overline{1} \rangle$ does not work, as $\langle e \rangle = \{e\}$. Keeping in mind that we are now working with

modulo 8, we have

$$\begin{aligned}
\langle\overline{3}\rangle &= \left\{\overline{3}, \overline{3}^2 = \overline{9} = \overline{1}\right\} \\
&= \left\{\overline{1}, \overline{3}\right\} \\
\langle\overline{5}\rangle &= \left\{\overline{5}, \overline{5}^2 = \overline{25} = \overline{1}\right\} \\
&= \left\{\overline{1}, \overline{5}\right\} \\
\langle\overline{7}\rangle &= \left\{\overline{7}, \overline{7}^2 = \overline{49} = \overline{1}\right\} \\
&= \left\{\overline{1}, \overline{7}\right\}.
\end{aligned}$$

We have just shown that $U(8)$ is *non-cyclic*.

One useful result for determining whether a group is cyclic or not is the following.

Remark 5. The group $U(n)$ is cyclic if $n = p^k$ or $n = 2p^k$, where p is an odd prime and $k \in \mathbb{N}$.

With these examples in mind, we can now explore cyclic groups and their properties in a more abstract way.

Theorem 4.1 (Criterion for $\mathbf{a}^i = \mathbf{a}^j$). Let G be a group and $a \in G$.

1. If $|a| = \infty$, then $a^i = a^j$ if and only if $i = j$.
2. If $|a| = n$, then $a^i = a^j$ if and only if $n \mid i - j$.

Proof.

1. Assume $|a| = \infty$. Then there is no nonzero integer n such that $a^n = e$, i.e., the sequence

$$\ldots, a^{-1}, e, a^1, a^2, \ldots,$$

never repeats. Thus, for $i, j \in \mathbb{Z}$, we have that $a^i = a^j$ if and only if

$$a^{i-j} = e \iff i - j = 0 \iff i = j.$$

2. Assume $|a| = n \in \mathbb{N}$. Thus, n is the smallest positive integer such that $a^n = e$. We want to show that $\langle a\rangle = \{e, a^1, \ldots, a^{n-1}\}$. Note that $\langle a\rangle = \left\{a^k \mid k \in \mathbb{Z}\right\}$. Let $k \in \mathbb{Z}$. Since $n \in \mathbb{N}$, by the division algorithm, there are $q \in \mathbb{Z}$ and $r \in \mathbb{Z}$ such that $k = qn + r$ with $0 \le r \le n - 1$. Then

$$\begin{aligned}
a^k &= a^{qn+r} \\
&= a^{qn}a^r \\
&= (a^n)^q a^r \\
&= (e)^q a^r \\
&= ea^r \\
&= a^r.
\end{aligned}$$

We thus have that $a^k \in \{e, a^1, \ldots, a^{n-1}\}$. Therefore, $\langle a \rangle = \{e, a^1, a^2, \ldots, a^{n-1}\}$. Assume that $a^i = a^j$. Then $a^{i-j} = a^{j-j} = a^0 = e$. By the division algorithm. there are integers q and r satisfying $i - j = qn + r$ with $0 \le r \le n-1$. Then $a^{i-j} = a^{qn+r}$. This implies that

$$\begin{aligned} e &= a^{qn+r} \\ &= a^{qn} a^r \\ &= a^r. \end{aligned}$$

Thus, $a^r = e$ and $0 \le r \le n-1$. Since n is the smallest natural integer such that $a^n = e$, we have $r = 0$. Thus, $i - j = qn$, i.e., $n \mid i - j$. To show the converse, assume $n \mid i - j$. Then, there is $q \in \mathbb{Z}$ such that $i - j = qn$. Thus,

$$\begin{aligned} a^{i-j} &= a^{qn} \\ &= (a^n)^q \\ &= e^q \\ &= e. \end{aligned}$$

□

Theorem 4.1 actually reveals the reason for the dual use of notation and terminology for the order of an element and the order of a group.

Corollary 1. For any group element a, $|a| = |\langle a \rangle|$.

Another special case of Theorem 4.1 occurs so often it deserves being singled out.

Corollary 2. Let G be a group and let a be an element of order n in G. If $a^k = e$, then $n \mid k$.

Another result of Theorem 4.1 is the relationship between $|ab|$ and $|a||b|$.

Corollary 3. If $a, b \in G$ and $ab = ba$, then $|ab| \mid |a||b|$.

Theorem 4.1 states that in the finite case, multiplication in $\langle a \rangle$ is essentially done by *addition* modulo n. Therefore, regardless what group we are working with, or how an element a in that group is chosen, multiplication in $\langle a \rangle$ works the same as addition in $\mathbb{Z}_n$ whenever $|a| = n$. Due to this, the cyclic groups $\mathbb{Z}$ and $\mathbb{Z}_n$ act as *prototypes* for all cyclic groups. We will explore this in-depth in Chapter 6.

The next theorem provides a method for computing $|a^k|$ when knowing only $|a|$.

Theorem 4.2. Let a be an element of order n in a group and let $k \in \mathbb{Z}^+$. Then

1. $\langle a^k \rangle = \langle a^{\gcd(n,k)} \rangle$.
2. $|a^k| = \frac{n}{\gcd(n,k)}$

Proof. Let $d = \gcd(n, k)$, $k = dr$ and $|a| = n$.

1. Since $a^k = a^{dr}$, we have $\langle a^k \rangle \subseteq \langle a^d \rangle$. By Bezout's theorem, there are integers s and t

such that $d = ns + kt$. Thus

$$\begin{aligned} a^d &= a^{ns+kt} \\ &= a^{ns}a^{kt} \\ &= (a^n)^s(a^k)^t \\ &= e^s(a^k)^t \\ &= (a^k)^t \in \langle a^k \rangle. \end{aligned}$$

Therefore, $\langle a^k \rangle = \langle a^d \rangle$.

2. We want to show that $|a^c| = \frac{n}{c}$ for any divisor c of n. Observe that $(a^c)^{n/c} = a^n = e$. Therefore, we have that

$$|a^c| \leq \frac{n}{c}.$$

Since $|a| = n$, we have $(a^c)^i \neq e$ for any $i \in \mathbb{N}$ such that $i < \frac{n}{c}$. This is because $ci < c \cdot \frac{n}{c} = n$. Therefore, $\frac{n}{c} \in \mathbb{N}$ is the smallest positive integer such that $(a^c)^{n/c} = e$. Let $c = \gcd(k, n)$. Then

$$\begin{aligned} |a^k| &= |\langle a^{\gcd(n,k)} \rangle| \\ &= \frac{n}{\gcd(n,k)} \end{aligned}$$

□

Let's illustrate this with an example.

Example 21. Suppose that $|a| = 36$ for $a \in G$. What is $|\langle a^8 \rangle|$? We have that $n = 36$ and $k = 8$. By the above theorem

$$\begin{aligned} |\langle a^8 \rangle| &= \frac{n}{\gcd(n,k)} \\ &= \frac{36}{\gcd(36,8)} \\ &= \frac{36}{4} \\ &= 9. \end{aligned}$$

Theorem 4.2 establishes an important relationship between the order of an element in a finite cyclic group and the order of the group itself.

Corollary 4. Let G be a finite cyclic group. For $g \in G$, $|g| \mid |G|$.

Corollary 5. Let $|a| = n$. Then $\langle a^i \rangle = \langle a^j \rangle$ if and only if $\gcd(n, i) = \gcd(n, j)$ and $|a^i| = |a^j|$ if and only if $\gcd(n, i) = \gcd(n, j)$.

The next two corollaries are special cases of the previous.

Corollary 6. Let $|a| = n$. Then $\langle a \rangle = \langle a^j \rangle$ if and only if $\gcd(n, j) = 1$ and $|a| = |a^j|$ if and only if $\gcd(n, j) = 1$.

Corollary 7. $\mathbb{Z}_n = \langle \overline{k} \rangle$ if and only if $\gcd(n, k) = 1$.

Keep in mind that Theorem 4.2 and its corollaries apply only to elements of finite orders.

4.1 Classification of Subgroups of Cyclic Groups

The next theorem tells us how many subgroups a finite cyclic group has and how to find them.

Theorem 4.3 (Fundamental Theorem of Cyclic Groups). Every subgroup of a cyclic group is cyclic. Moreover, if $|\langle a \rangle| = n$, then the order of any subgroup of $\langle a \rangle$ is a divisor of n; and, for each positive divisor k of n, the group $\langle a \rangle$ has exactly one subgroup of order k – namely, $\langle a^{n/k} \rangle$.

Proof. Let $G = \langle a \rangle$ and $H \leq G$. If $H = \{e\}$, then $H = \langle a^n \rangle = \langle e \rangle = \{e\}$. Suppose $H \neq \{e\}$. Since $H \leq G = \langle a \rangle$, every element of H is of the form a^t for some $t \in \mathbb{Z}$. Let $S = \{k \in \mathbb{N} \mid a^k \in H\}$. Note that $a^{-t} \in H$ because H is a group. Since $H \neq \{e\}$, we have that $H \neq \emptyset$. By the Well Ordering Principle, S has the smallest element, say $m \in S$. Since $a^m \in H$, $\langle a^m \rangle \subseteq H$. We claim that $H = \langle a^m \rangle$. It suffices to show that $H \subseteq \langle a^m \rangle$. Let $b \in H$. Thus, $b \in G$. Since G is cyclic, $b = a^k$ for some $k \in \mathbb{Z}$. By the division algorithm, there are q and r such that $k = qm + r$ with $0 \leq r \leq m - 1$. Then $a^k = a^{qm} a^r$. Thus, $a^{-qm} a^k = a^r \in H$. Since m is the least positive integer such that $a^m \in H$ and $0 \leq r < m$, we have that $r = 0$. Therefore, $b = a^k = a^{mq} = (a^m)^q \in \langle a^m \rangle$. We can conclude that $H \subseteq \langle a^m \rangle$. Hence, $H = \langle a^m \rangle$.

To show the second part, let $|G| = |\langle a \rangle| = n$ and $k \in \mathbb{N}$ such that $k \mid n$. Now we will show that there is a unique subgroup of G with order k. To show existence, choose $H = \langle a^m \rangle$ where $m = \gcd(n, m)$. This is because, we have that

$$|\langle a^m \rangle| = \frac{n}{\gcd(m, n)} = k.$$

Let's call $l = \gcd(m, n)$. We have that $n = kl, k = \frac{n}{l}$, and $l = \frac{n}{k}$. Therefore, we choose $m = l$. Thus

$$\begin{aligned} |H| &= |\langle a^m \rangle| \\ &= \frac{n}{\gcd(m, n)} \\ &= \frac{n}{m} \\ &= \frac{n}{\frac{n}{k}} \\ &= k. \end{aligned}$$

Next we will show uniqueness. Assume that we have two subgroups $H_1 = \langle a^{m_1} \rangle$ and $H_2 = \langle a^{m_2} \rangle$ such that $|H_1| = |H_2|$ where $m_1, m_2 \in \mathbb{N}$ are the smallest members for H_1 and H_2,

respectively. We have that

$$\begin{aligned} |H_1| &= |\langle a^m \rangle| \\ &= \frac{n}{\gcd(m_1, n)} && (1) \\ |H_2| &= \frac{n}{\gcd(m_2, n)} && (2) \end{aligned}$$

Equating (1) and (2), we have that

$$\frac{n}{\gcd(m_1, n)} = \frac{n}{\gcd(m_2, n)} \iff \gcd(m_1, n) = \gcd(m_2, n).$$

Thus, $m_1 = m_2$. □

What does this mean? Suppose $G = \langle a \rangle$ and $|G| = 30$. The first and second parts of the theorem say that if $H \leq G$, then H has the form $\langle a^{30/k} \rangle$ for some k that is a divisor of 30. The third part of the theorem says that G has one subgroup of each of the orders $1, 2, 3, 5, 6, 10, 15$, and 30, with no others. Taking the group in Theorem 4.3 to me $\mathbb{Z}_n$ and a to be 1, we obtain the following important special case.

Corollary 8 (Subgroups of $\mathbb{Z}_n$). For each positive divisor k of n, the set $\langle n/k \rangle$ is the unique subgroup of $\mathbb{Z}_n$ of order k; moreover, these are the only subgroups of $\mathbb{Z}_n$.

Combining Theorems 4.2 and 4.3, we can easily count the number of elements of each order in a finite cyclic group. To aid us with this, we introduce the a function important in number theory: the *Euler phi function*.

Definition (Euler Phi Function). Let $n \in \mathbb{N}$. The Euler phi function $\phi(n)$ is the number of coprimes i of n such that $1 \leq i \leq n$.

Example 22. For $n = 1$, if $\gcd(i, n) = 1$ for $1 \leq i < 1$, then $i = 1$ and $\phi(1) = 1$. For $n = 2$, $\gcd(i, 2) = 1, 1 \leq i \leq 2$. $\phi(2) = 1$ as well. For $n = 3$, $\gcd(i, 3) = 1, 1 \leq i \leq 3$, $\phi(3) = 2$. For a prime p, $\phi(p) = p - 1$.

We can use the Euler phi function as follows.

Theorem 4.4 (Number of Elements of Each Order in a Cyclic Group). If d is a positive divisor of n, the number of elements of order d in the cyclic group with order n is $\phi(d)$.

Although there isn't a formula for the number of elements of each order for arbitrary finite groups, we still have the following corollary that can be useful.

Corollary 9 (Number of Elements of Order d in a Finite Group). In a finite group, the number of elements of order d is a multiple of $\phi(d)$.

Although cyclic groups constitute a very small class of finite groups, we will see in Chapter 11 that they act as building blocks for all finite Abelian groups, in much the same way that primes are the building blocks for the integers.

Chapter 5

Permutation Groups

In this chapter, we will study a particular group of functions called *permutation groups*. In the early and mid-19th century, these were the only groups investigated by mathematicians. It was not until 1850 that Cayley introduced the concept of an abstract group (and took even longer for the idea to firmly take hold).

Definition (Permutation). Let A be a set. A function $f : A \to A$ that is one-to-one and onto (bijective) is called a *permutation* on A.

Example 23. Consider $A = \{a\}$. The only possible permutation is $f(a) = a$. What about $A = \{a, b\}$? We could have the permutation

$$\begin{aligned} f(a) &= a \\ f(b) &= b, \end{aligned}$$

or the permutation

$$\begin{aligned} f(a) &= b \\ f(b) &= a. \end{aligned}$$

What about something like $f(a) = f(b) = a$? This is not one-to-one/onto. Therefore, we can conclude that there are 2 bijective functions on A.

We call the mapping $f(x) = x, \forall x \in A$ the *identity*, denoted ε.

Example 24. What if $A = \{a, b, c\}$? How many permutations are there? The number of permutations is $|A|! = 3! = 6$.

As a fun exercise, we can quickly show why $0! = 1$.

Example 25. Suppose $A = \emptyset$. Does there exist a function $f : \emptyset \to \emptyset$? Generally, we define a function from A to B as $\forall a \in A, \exists! b \in B$ such that $(a, b) \in B$. In English, we have "If $a \in A$, then there is $b \in B$ such that $(a, b) \in B$". Since the statement $a \in A$ is always false for $\emptyset$, the statement is always true. Therefore, there is a function $f : \emptyset \to \emptyset$. This function is $f_\emptyset = \emptyset$,

which is a permutation on $A = \emptyset$. In other terms, the permutations on $\emptyset$ is $\{f_\emptyset\} = \{\emptyset\}$. Thus, the permutations on $\emptyset$ is 1. Thus, $0! = 1$.

A more convenient way to denote permutations is to write them in *array* form. For example, we could define a permutation α of the set $\{1,2,3,4\}$ by specifying

$$\alpha(1) = 2, \quad \alpha(2) = 3, \quad \alpha(3) = 1, \quad \alpha(4) = 4.$$

In array form, we would write this like

$$\alpha = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 1 & 4 \end{bmatrix}.$$

Observe that $\alpha(i)$ is placed directly below i for each i. We can also compose permutations, consider the example below.

Example 26. Suppose $A = \{1,2,3,4\}$ and G is a group of permutations (one-to-one and onto) on A. Let

$$\alpha = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \end{bmatrix}, \quad \beta = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 3 & 2 & 1 & 4 \end{bmatrix}.$$

The product/composition of α and β, denoted $\alpha\beta$ or $\alpha \circ \beta$, can be performed by working right to left. For example, consider $\alpha\beta(3)$.

$$\alpha\beta(3) = \alpha(\beta(3)) = \alpha(1) = 2.$$

Continuing this process for the rest of the values, we obtain

$$\alpha\beta = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \end{bmatrix}.$$

Let's define some standard notation that we'll use from now on.

Definition (Symmetric Group). Let $A = \{1,2,\ldots,n\}$. The set of all permutations of A is called the *symmetric group of degree n*, denoted S_n.

Example 27. Let's take a look at the symmetric group S_3. Of course it has the identity permutation

$$\varepsilon = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \end{bmatrix}.$$

We can also define following permutation and find its square

$$\alpha = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{bmatrix}, \quad \alpha^2 = \begin{bmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{bmatrix}.$$

Let's also consider the permutation

$$\beta = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 3 & 2 \end{bmatrix}.$$

Composing α and β, we obtain

$$\alpha\beta = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{bmatrix}, \quad \beta\alpha = \begin{bmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{bmatrix}.$$

> We've actually just shown that S_3 is non-abelian, as $\alpha\beta \neq \beta\alpha$.

It can actually be proven that S_n is non-abelian for $n \geq 3$.

5.1 Cycle Notation

Another notation commonly used to specify permutations is called *cycle notation*. It was first introduced by Cauchy in 1815. Consider the following permutation of S_6

$$\alpha = \begin{bmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 2 & 1 & 4 & 5 & 3 & 6 \end{bmatrix}$$

We observe that the elements of this permutation have cycles, which could be presented as follows

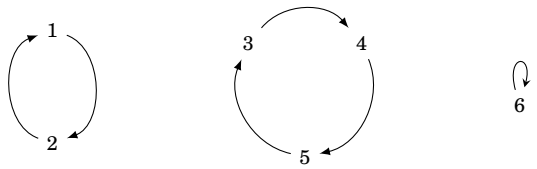


Figure 5.1: Disjoint cycles of α

We can actually represent a permutation by its disjoint cycles, i.e., in this case we'd write α as (12)(345)(6). Let's write a pseudo-formal definition for what an m-cycle actually is.

Definition. Let $\alpha \in S_n$ with $a_i \in \{1, 2, \ldots, n\}$ and

$$\alpha(a_1) = a_2, \quad \alpha(a_2) = a_3, \quad \ldots, \quad \alpha(a_{m-1}) = a_{m-1}, \quad \alpha(a_m) = a_1.$$

α is called an m-cycle of S_n. We write $\alpha = (a_1 a_2 \ldots a_m)$.

We also see that these disjoint cycles are abelian, that is to say

$$\begin{aligned} \alpha &= (12)(345)(6) \\ &= (345)(12)(6) \\ &= (6)(12)(345) \\ &\vdots \end{aligned}$$

We now provide a more rigorous definition for disjoint cycles.

Definition (Disjoint Cycles). Let $\alpha, \beta \in S_n$ be the cycles $\alpha = (a_1 a_2 \ldots a_n)$ and $\beta = (b_1 b_2 \ldots b_m)$. α and β are *disjoint cycles* if

$$\{a_1, a_2, \ldots, a_n\} \cap \{b_1, b_2, \ldots, b_m\} = \emptyset.$$

We also remark the commutative property of disjoint cycles.

Remark 6. $\alpha\beta = \beta\alpha$ for any disjoint cycles α and β.

We can multiply cycles by thinking of them as permutations given in array form. Consider $\alpha = (13)(27)(456)(8)$ and $\beta = (1237)(648)(5)$. We could say that

$$\alpha\beta = (13)(27)(456)(8)(1237)(648)(5),$$

however, it is usually desirable to express a permutation is *disjoint* cycle form. Keeping in mind that function composition is done from right to left and that each cycle that does not contain an element fixes the element to itself, we observe that: (5) fixes 1; (648) fixes 1; (1237) sends 1 to 2; (8) fixes 2; (456) fixes 2; (27) sends 2 to 7; and (13) fixes 7. So the net effect of $\alpha\beta$ is to send 1 to 7. Thus, we can begin with $\alpha\beta = (17\ldots)\ldots$. Now, we need merely repeat the process for the remaining elements.

5.2 Properties of Permutations

We are now ready to state several theorems about permutations and cycles.

Theorem 5.1 (Products of Disjoint Cycles). Every permutation of a finite set can be written as a cycle or as a product of disjoint cycles.

Proof. Let $\alpha \in S_n$ for $n \in \mathbb{N}$. In other terms, α is a permutation on $A = \{1,2,\ldots,n\}$. Choose any member of A, say a_1 and let $a_2 = \alpha(a_1), a_3 = \alpha(a_2)$, and so on until we arrive at $a_1 = \alpha^m(a_1)$ for some m. We know that m exists since A is finite, and there must eventually be a repetition. If $\{a_1,a_2,\ldots,a_m\} = \{1,2,\ldots n\}$, we are done. It it is not the case, we continue to choose $b_1 \in \{1_2,\ldots,n\}\setminus\{a_1,a_2,\ldots,a_m\}$, i.e., considering an element in the set with the first disjoint cycle excluded. We will thus have $\alpha^k(b_1) = b_1$ for some $k \in \mathbb{N}$. This process will end in at most n times, as we have n elements to work with. Thus, we can write α as a product of disjoint cycles

$$\alpha = (a_1a_2\ldots a_m)(b_1b_2\ldots b_k)\ldots$$

□

From the theorem above, we see that every permutation can be written as a product of disjoint cycles.

Theorem 5.2 (Disjoint Cycles Commute). If α and β are disjoint cycles in S_n, then $\alpha\beta = \beta\alpha$.

So far we've seen that writing permutations in disjoint cycle form is a little more efficient, but there are more advantages to doing so. The next theorem (Ruffini, 1799) shows us that the disjoint cycle form allows us to "eyeball" the order of a permutation.

Theorem 5.3 (Order of a Permutation). The order of a permutation of a finite set written in disjoint cycle form is the least common multiple of the lengths of the cycle.

Proof. Let $\alpha \in S_n$ for $n \in \mathbb{N}$. By Theorem 5.1, α can be written as $\alpha = \alpha_1\alpha_2\ldots\alpha_k$ where α_i's are a (mutually) disjoint cycle. By Theorem 5.2, $\alpha^m = \alpha_1^m\alpha_2^m\ldots\alpha_k^m$ for $m \in \mathbb{N}$. Let $|\alpha_i| = m_i$

for $i = 1, 2, \ldots, k$. Let $m^* = \text{lcm}(m_1, m_2, \ldots, m_k)$. Then

$$\begin{aligned} \alpha^{m^*} &= \alpha_1^{m^*} \alpha_2^{m^*} \ldots \alpha_k^{m^*} \\ &= \varepsilon \cdot \varepsilon \ldots \varepsilon \\ &= \varepsilon. \end{aligned}$$

Moreover, $m^* \in \mathbb{N}$ is the minimum positive integer such that $\alpha^{m^*} = \varepsilon$. □

This is a tremendous advantage, as the general method for finding the order of a permutation is quite tedious and computationally expensive. We see in the example below just how easy it is to apply Theorem 5.3.

Example 28. The order of the permutation (123)(45)(6) is

$$|(123)(45)(6)| = \text{lcm}(3, 2, 1) = 6.$$

As we will soon see, it is often greatly advantageous to write a permutation as a product of cycles of length two, i.e., permutations of the form (ab) where $a \neq b$. These permutations are often called *transpositions*, as the effect of (ab) is to transpose/interchange a and b.

Theorem 5.4 (Product of 2-Cycles). Every permutation in S_n, for $n > 1$, is a product of 2-cycles.

Proof. Let $\alpha \in S_n$ where $n \in \mathbb{N}$ and $n > 1$. By Theorem 5.1, α can be written as the product of disjoint cycles $\alpha_1 \alpha_2 \ldots \alpha_k$. For $i \in \{1, \ldots, k\}$, write

$$\begin{aligned} \alpha_i &= (a_1 a_2 \ldots a_m) \\ &= (a_1 a_m)(a_1 a_{m-1}) \ldots (a_1 a_2). \end{aligned}$$

Thus, any disjoint cycle α_i can be written as a product of 2-cycles. Therefore, $\alpha = \alpha_1 \alpha_2 \ldots \alpha_k$ itself can be written as a product of 2-cycles. □

We observe that a permutation can be written in terms of an even or odd amount of cycles.

Lemma 1. Let $\varepsilon = \beta_1 \beta_2 \ldots \beta_r$ where B_i's are 2-cycles. Then r is even.

Proof (Book). Note that $r \neq 1$. We will prove by induction on r. If $r = 2$, we are done. Assume that $\varepsilon = \beta_1 \beta_2 \ldots \beta_k$, where β_i's are 2-cycles and $k \leq r$. Then k is even. Assume that $\varepsilon = \alpha_1 \alpha_2 \ldots \alpha_{r+1}$, where α_i's are 2-cycles. Suppose that the rightmost 2-cycle is (ab). Then since $(ij) = (ji)$, the product $\beta_{r-1}\beta_r$ can be expressed in one of the following forms shown on the right:

$$\begin{aligned} \varepsilon &= (ab)(ab) \\ (ab)(bc) &= (ac)(ab) \\ (ac)(cb) &= (bc)(ab) \\ (ab)(cd) &= (cd)(ab). \end{aligned}$$

If the first case occurs, we may delete $\beta_{r-1}\beta_r$ from the original product to obtain $\varepsilon = \beta_1 \beta_2 \ldots \beta_{r-2}$,

and therefore, by the mathematical hypothesis, $r-2$ is even. In the other 3 cases, consider $\alpha_{r+1}\alpha_r = (ac)(cb)$.

In the other three cases, we replace the form of $\beta_{r-1}\beta_r$ on the right by its counterpart on the left to obtain a new product of r 2-cycles that is still the identity, but where the rightmost occurrence of the integer a is in the second-from-the-rightmost 2-cycle of the product instead of the rightmost 2-cycle. We now repeat the procedure just described with $\beta_{r-2}\beta_{r-1}$, and as before, we obtain a produce of $(r-2)$ 2-cycles equal to the identity or a new product of r 2-cycles, where the rightmost occurrence of a is in the third 2-cycle from the right. Continuing this process, we must obtain a product of $(r-2)$ 2-cycles equal to the identity, because otherwise we have a product equal to the identity in which the only occurrence of the integer a is in the leftmost 2-cycle, and such a product does not fix a, whereas the identity does. Hence, by mathematical induction, $r-2$ is even, and r is even as well. □

Proof (Class). Let $\varepsilon \neq (ab) \in S_n$, since any 2-cycle/transposition is not the identity. The length of ε is thus at least 2, i.e., $\epsilon = \beta_1\beta_2\ldots\beta_n$, where β_i's are 2-cycles, implies $n \geq 2$. We will show that ε is an even permutation. We will continue by induction on n. Write $\varepsilon = \beta_1\beta_2\ldots\beta_r$, where $r \geq 2$.

<u>Base case</u>

For $r = 2$, we are done, since r is even.

<u>Inductive step</u>

Assume that if $\varepsilon = \alpha_1\alpha_2\ldots\alpha_k$ for $2 \leq k \leq r$ where α_i's are 2-cycles, then k is even. If $\alpha_{r+1} = (ab)$, then we have that $\alpha_r\alpha_{r+1}$ could be

$$\alpha_r\alpha_{r+1} = \begin{cases} (ab)(ab) = \varepsilon \\ (ac)(ab) = (abc) = (ab)(bc) \\ (bc)(ab) = (acb) = (ac)(cb) \\ (cd)(ab) = (ab)(cd) \end{cases}$$

Note how we rewrote to have a on the left hand side for each case. Continuing this algorithm, we will have one of the two cases

1. a is on the left-most of the product, i.e., $(ax)(yz)$.
2. a is gone.

But which one will we end up with? It turns out, rather unintuitively, that we end up with the second. If we were to end the algorithm with the first case, then $\varepsilon(a) = x \neq a$, which is a contradiction (i.e., a does not map to itself so we cannot have ε). Thus, a must disappear in a finite number of steps. Therefore, $\varepsilon = \gamma_1\gamma_2\ldots\gamma_{r-1}$, where γ_j's are 2-cycles. By the inductive hypothesis, since $r-1 \leq r$, $k = r-1$ is even.

By mathematical induction, ε is an even permutation. □

Theorem 5.5 (Always Even or Always Odd). If a permutation α can be expressed as a product of an even (odd) number of 2-cycles, then every decomposition of α into a product of 2-cycles must have an even (odd) number of 2-cycles. In symbols, if

$$\alpha = \beta_1\beta_2\ldots\beta_r, \quad \alpha = \gamma_1\gamma_2\ldots\gamma_s,$$

where the β's and the γ's are 2-cycles, then r and s are both even or both odd.

Proof. Let $\alpha \in S_n$. Assume that $\alpha = \beta_1\beta_2\ldots\beta_r = \gamma_1\gamma_2\ldots\gamma_s$, where β_i's and γ_j's are 2-cycles. We have that

$$\begin{aligned}\varepsilon &= \alpha\alpha^{-1}\\ &= \beta_1\beta_2\ldots\beta_r\gamma_s^{-1}\gamma_{s-1}\ldots\gamma_1^{-1}\\ &= \beta_1\beta_2\ldots\beta_r\gamma_s\gamma_{s-1}\ldots\gamma_1.\end{aligned}$$

By the previous lemma, $r+s$ is even. Thus, $r-s = r+s-2s$ is also even. □

Theorems 5.4 and 5.5 together show that every permutation can be unambiguously classified as either even or odd. The significance of this observation is given in Theorem 5.6.

Theorem 5.6 (Even Permutations Form a Group). The set of even permutations in S_n forms a subgroup of S_n.

Proof. We will show this is true for $n \geq 2$. We know that $A_n \neq \emptyset$, as ε is an even permutation, and therefore $\varepsilon \in A_n$. Next we will show closure, i.e., $\alpha, \beta \in A_n$ implies $\alpha\beta \in A_n$. Multiplying α and β, both even cycles, we obtain another even cycle. Therefore, A_n is closed. For inverses, we have that for $\alpha \in A_n$, all we are doing is reversing the cycle, so $\alpha^{-1} \in A_n$. Therefore, the set of even permutations in S_n forms a subgroup of S_n. □

The subgroup of even permutations in S_n arises so often that we give it a special name and notation.

Definition (Alternating Group of Degree n). The group of even permutations of n symbols is denoted by A_n and is called the alternating group of degree n.

Furthermore, we also note that half the elements of S_n are even permutations (for $n > 1$).

Theorem 5.7. For $n > 1$, A_n has order $\frac{n!}{2}$.

Proof. For each odd permutation α, the permutation $(12)\alpha$ is even and by the cancellation property in groups, $(12)\alpha \neq (12)\beta$, when $\alpha \neq \beta$. Thus, there are at least as many even permutations as there are odd ones. One the other hand, for each permutation α, the permutation $(12)\alpha$ is odd and $(12)\alpha \neq (12)\beta$ when $\alpha \neq \beta$. Thus, there are at least as many odd permutations as there are even ones. If follows that there are an equal number of even and odd permutations. Since $|S_n| = n!$, we have that $|A_n| = n!/2$. □

Before moving onto chapter 6, let's show how we can use even and odd permutations to define the determinant operation of a matrix.

Definition. Let $\sigma \in S_n$. We define

$$\operatorname{sign}(\sigma) = \begin{cases} +1, & \sigma \text{ is an even permutation} \\ -1, & \sigma \text{ is an odd permutation} \end{cases}.$$

Let $A = [a_{ij}]_{n\times n}$. The determinant of A is

$$\det(A) = \sum_{\sigma\in S_n} \operatorname{sign}(\sigma) a_{1\sigma(1)} a_{2\sigma(2)} \dots a_{n\sigma(n)},$$

where $a_{i\sigma(i)}$ is equivalent to a_{ij}.

Example 29. Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$. We let $S_2 = \{\varepsilon, (12)\}$. We have that $\operatorname{sign}(\varepsilon) = 1$ and $\operatorname{sign}((12)) = -1$. Therefore,

$$\begin{aligned}
\det(A) &= \sum_{\sigma\in S_2} \operatorname{sign}(\sigma) a_{1\sigma(1)} a_{2\sigma(2)} \\
&= \operatorname{sign}(\varepsilon) a_{11} a_{22} + \operatorname{sign}((12)) a_{12} a_{21} \\
&= (+1)ad + (-1)bc \\
&= ad - bc.
\end{aligned}$$

Chapter 6

Isomorphisms

6.1 Motivation

In this chapter, we give a formal method for determining whether two groups defined in different terms are really the same. When this is the case, we say that there an *isomorphism* between the two groups*. Analogously, suppose an American and a Thai are asked to count objects. The American says "One, two, three", whereas the Thai says "*Nueng, song, same*". the two are doing the same thing but are using different terminology to do so. The two are in agreement on the *concept* they are describing, even if they aren't speaking the same language.

Example 30. Consider the group

$$(\mathbb{Z}_4, +) = \{\overline{0}, \overline{1}, \overline{2}, \overline{3}\}.$$

Another group with practically the same elements is

$$(U(5), \cdot) = \{\overline{1}, \overline{2}, \overline{3}, \overline{4}\}.$$

Naturally, the two are not the same group. However, let's examine their structure. In $\mathbb{Z}_4$, we have

$$\begin{aligned} \overline{1} + \overline{1} &= \overline{2} \\ \overline{1} + \overline{1} + \overline{1} &= \overline{3} \\ \overline{1} + \overline{1} + \overline{1} + \overline{1} &= \overline{0}. \end{aligned}$$

On the other hand, for $U(5)$, we have

$$\begin{aligned} \overline{2} \cdot \overline{2} &= \overline{4} \\ \overline{2} \cdot \overline{2} \cdot \overline{2} &= \overline{3} \\ \overline{2} \cdot \overline{2} \cdot \overline{2} \cdot \overline{2} &= \overline{1}. \end{aligned}$$

$(\mathbb{Z}_4, +)$ is "similar" to $(U(5), \cdot)$ in the sense that their identity and elements match.

*The term *isomorphism* was coined by Galois himself and is derived from the Greek words *isos*, meaning "same" or "equal", and *morphe*, meaning "form".

With this in mind, we introduce the main concept for this chapter: *isomorphisms*.

6.2 Definition and Examples

Definition (Group Isomorphism). Let $(G,\cdot)$ and $(\overline{G},*)$ be groups. An *isomorphism* ϕ from G to $\overline{G}$ is a one-to-one mapping (or function) from G onto $\overline{G}$ that preserves the group operation. That is,

$$\phi(a\cdot b)=\phi(a)*\phi(b), \quad \forall a,b\in G.$$

If there is an isomorphism from G onto $\overline{G}$, we say that G and $\overline{G}$ are *isomorphic* and write $G\approx\overline{G}$.

It is implicit in the definition of an isomorphism that isomorphic groups have the same order. It is also implicit that the operation of the left hand side of the equal sign is that of G, whereas the operation on the right belongs to $\overline{G}$. There are four separate steps involved in proving that a group G is isomorphic to a group $\overline{G}$.

1. **Mapping:** define a candidate for the isomorphism; that is, define a function $\phi: G\to\overline{G}$.
2. **Injectivity:** prove that ϕ is one-to-one; that is, assume $\phi(a)=\phi(b)$ and prove that $a=b$, where $a,b\in G$.
3. **Surjectivity:** prove that ϕ is onto; that is, for any element $\overline{g}\in\overline{G}$, it is possible to find an element $g\in G$ such that $\phi(g)=\overline{g}$.
4. **Operation Preserving:** prove that ϕ is operation preserving; that is, show that $\phi(ab)=\phi(a)\phi(b)$ for all $a,b\in G$.

With that in mind, let's look at some examples.

Example 31. Let $G=(\mathbb{Z}_4,+)$ and $\overline{G}=(U(5),\cdot)$. We have that $\phi(\overline{1})=\overline{2}$. For $\overline{i}\in\mathbb{Z}_4$, we define

$$\phi(\overline{i})=\left(\overline{2}\right)^i.$$

Example 32. Let $G=(\mathbb{R},+)$ and $\overline{G}=(\mathbb{R}^+,\cdot)$. Define $\phi: G\to\overline{G}$ by $\phi(x)=e^x$ for $x\in\mathbb{R}$.

To verify that ϕ is one-to-one, let $a,b\in\mathbb{R}$ such that $\phi(a)=\phi(b)$. Thus, $e^a=e^b$. Since the exponential function is increasing on $\mathbb{R}$, ϕ is one-to-one. Alternatively, we could show this by taking the natural log on both sides.

To show that ϕ is onto, let $y\in\mathbb{R}^+$. Choose $x=\ln y$. Then $e^x=e^{\ln y}=y$.

Finally, to show that ϕ is operation-preserving, let $a,b\in\mathbb{R}$. We have that

$$\begin{aligned}\phi(a+b)&=e^{a+b}\\&=e^ae^b\\&=\phi(a)\cdot\phi(b).\end{aligned}$$

Hence, we can conclude that $G=(\mathbb{R},+)$ and $\overline{G}=(\mathbb{R}^+,\cdot)$ are isomorphic.

Example 33. Let $G = S_3$, where

$$S_3 = \{\varepsilon, (12), (13), (23), (123), (132)\}.$$

We will try to create an isomorphism from G to itself. For all $\alpha \in S_3$, let $\phi(\alpha) = (123)\alpha$. Is ϕ an isomorphism? Let's find out. Assume that $\phi(\alpha_1) = \phi(\alpha_2)$, i.e.,

$$(123)\alpha_1 = (123)\alpha_2.$$

We can use cancellation/multiply by inverse to obtain $\alpha_1 = \alpha_2$. Therefore, ϕ is one-to-one. Next, let $\beta \in S_3$. Choose $\alpha_0 = (123)^{-1}\beta$. Then, we have that

$$\phi(\alpha_0) = (123)\left[(123)^{-1}\beta\right] = \beta.$$

Finally, let $\tau_1, \tau_2 \in S_3$. We have that

$$\phi(\tau_1\tau_2) = (123)\tau_1\tau_2 \neq [(123)\tau_1][(123)\tau_2].$$

Therefore, ϕ is not an isomorphism from G to itself.

6.3 Cayley's Theorem

The function ϕ from the previous example was almost working, let's try to fix it. We have that $|S_3| = 6$. What if we tried creating an isomorphism from S_3 to S_6? It seems like Cayley had the same idea.

Theorem 6.1 (Cayley's Theorem). Every group is isomorphic to a subgroup of permutations. In other terms, any group $G \cong \overline{G}$, where $\overline{G} \leq S_G$.

Proof. We claim that $G \cong \overline{G}$, where

$$\overline{G} = \{\phi_g \in S_G \mid \phi_g(x) = gx \text{ for } x \in G\}$$

and furthermore

$$S_G := \{f : G \to G \mid f \text{ is bijective}\}.$$

For any $g \in G$, we define $\phi : G \to \overline{G}$ as follows: let $\phi(g) = \phi_g$ be a function on G where $\phi_g(x) = gx$ for any $x \in G$. We claim that this is an isomorphism. Why?

1) Let $g_1, g_2 \in G$ such that $\phi(g_1) = \phi(g_2)$. Thus, $\phi_{g_1} = \phi_{g_2}$, not as elements, but as functions. Choose $x = e \in G$.

$$\begin{aligned} \phi_{g_1}(e) &= \phi_{g_2}(e) \\ g_1 e &= g_2 e \\ g_1 &= g_2. \end{aligned}$$

   Therefore, ϕ is one-to-one.

2) Let $T \in \overline{G} \subseteq S_G$. Thus, there is $g_0 \in G$ such that $T(x) = g_0 x$ for $x \in G$. Choose $g = g_0$, then

$$\phi(g) = \phi_g(x) = gx = T(x) = T.$$

Therefore, ϕ is onto.

3) Finally, to show that ϕ is order-preserving (and also that $\overline{G} < S_G$), let $g_1, g_2 \in G$. We want to show that $\phi(g_1 g_2) = \phi(g_1) \circ \phi(g_2)$. Let $x \in G$. We have that

$$\begin{aligned}\phi(g_1 g_2)(x) &= \phi_{g_1 g_2}(x) \\ &= (g_1 g_2)x \\ &= g_1(g_2(x)) \\ &= g_1(\phi_{g_2}(x)) \\ &= (\phi_{g_1} \circ \phi_{g_2})(x).\end{aligned}$$

Thus, $\phi(g_1 g_2) = \phi_{g_1} \circ \phi_{g_2}$, hence ϕ is order-preserving.

We can thus conclude that ϕ is an isomorphism. □

6.4 Properties of Isomorphisms

Let's introduce another "trivial"[†] theorem.

Theorem 6.2 (Properties of Isomorphisms Acting on Elements). Assume that $G \cong \overline{G}$, where both groups contain identities e and $\overline{e}$, respectively. Suppose that ϕ is an isomorphism from G onto $\overline{G}$.

1. ϕ carries the identity of G to the identity of $\overline{g}$. That is to say, $\phi(e)$ is an identity in $\overline{G}$.
2. For every integer n and for every group element $a \in G$, $\phi(a^n) = [\phi(a)]^n$.
3. For any elements $a, b \in G$, a and b commute if and only if $\phi(a)$ and $\phi(b)$ commute.
4. $G = \langle a \rangle$ if and only if $\overline{G} = \langle \phi(a) \rangle$.
5. ϕ preserves order, which is to say, $|a| = |\phi(a)|$.
6. For a fixed integer k and a fixed group element $b \in G$, the equation $x^k = b$ has the same number of solutions in G as does the equation $x^k = \phi(b)$ in $\overline{G}$.
7. If G is finite, then G and $\overline{G}$ have exactly the same number of elements of every order (and thus $|G| = |\overline{G}|$).

Proof. We will prove only the first property. For (1), let $e \in G$ and $\overline{e} \in \overline{G}$ be identities in their respective groups. We want to show that for an isomorphism $\phi : G \to \overline{G}$, we have that $\phi(e) = \overline{e}$. Let $\overline{g} \in \overline{G}$. We know that ϕ is one-to-one, onto, and order preserving. Due to the fact that it is onto, there is $g \in G$ such that $\phi(g) = \overline{g}$. Therefore, we have

$$\begin{aligned}\phi(eg) &= \phi(g) \\ \phi(e)\phi(g) &= \phi(g) && \text{(operating preserving)} \\ \phi(e)\phi(g) &= \overline{e}\phi(g) \\ \phi(e) &= \overline{e} && \text{(cancellation)}\end{aligned}$$

□

[†] "I'm gonna use the T-word here" – Ajarn Chatchawan, 2023

Although the previous theorem seems quite technical, it merely comes from the fact that two isomorphic groups have the same structure. Next, we introduce a similar theorem.

Theorem 6.2 (Properties of Isomorphisms Acting on Groups). Suppose that ϕ is an isomorphism from a group G onto a group $\overline{G}$. Then

1. ϕ^{-1} is an isomorphism from $\overline{G}$ onto G.
2. G is Abelian if and only if $\overline{G}$ is Abelian.
3. G is cyclic if and only if $\overline{G}$ is cyclic.
4. If K is a subgroup of G, then $\phi(K) = \{\phi(k) \mid k \in K\}$ is a subgroup of $\overline{G}$.
5. If $\overline{K}$ is a subgroup of $\overline{G}$, then $\phi^{-1}(\overline{K}) = \{g \in G \mid \phi(g) \in \overline{K}\}$ is a subgroup of G.
6. $\phi(Z(G)) = Z(\overline{G})$.

Theorems 6.2 and 6.3 provide several convienient ways to prove that G and $\overline{G}$ are not isomorphic.

1. Observe that $|G| \neq |\overline{G}|$.
2. Observe that G or $\overline{G}$ is cyclic while the other is not.
3. Observe that G or $\overline{G}$ is Abelian and the other is not.
4. Show the the largest order of any element in G is not the same as the largest order of any element in $\overline{G}$.
5. Show that the number of elements of some specific order in G (the smallest order greater than 1 is often the good choice) is not the same as the number of elements of that order in $\overline{G}$.

6.5 Automorphisms

Certain kinds of isomorphisms are referred to so often that they have been given special names.

Definition (Automorphism). An isomorphism from a group G onto itself is called an *automorphism* of G.

The word *automorphism* comes from the Greek *auto* meaning "self" and *morph* meaning "shape".

Example 34. The identity mapping $i(a) = a$, for all a in a group G, is a 'trivial' automorphism.

Example 35. Let $G = GL_2(\mathbb{R})$, the set of 2×2 invertible matrices with real elements. Let $A \in G$. Define $\phi_A : G \to G$ by

$$\phi_A(M) = AMA^{-1}, \quad M \in G.$$

Since we have that

$$\begin{aligned}\det\big(\phi_A(M)\big) &= \det\big(AMA^{-1}\big)\\ &= \det(A)\det(M)\det\big(A^{-1}\big)\\ &\neq 0,\end{aligned}$$

$\phi_A(M)$ is still in G. To show that ϕ_A is one-to-one, let $M_1, M_2 \in G$ such that $\phi_A(M_1) = \phi_A(M_2)$. We have that

$$\begin{aligned} AM_1A^{-1} &= AM_2A^{-1} \\ A^{-1}(AM_1A^{-1})A &= A^{-1}(AM_2A^{-1})A \\ M_1 &= M_2. \end{aligned}$$

To show that ϕ_A is onto, let $M \in G$. Choose $N \in G$ such that $N = A^{-1}MA$.

$$\begin{aligned} \phi_A(N) &= ANA^{-1} \\ &= A(A^{-1}MA)A^{-1} \\ &= M. \end{aligned}$$

Finally, to show that ϕ is operation preserving, let $N_1, N_2 \in G$.

$$\begin{aligned} \phi_A(N_1N_2) &= A(N_1N_2)A^{-1} \\ &= A(N_1A^{-1}AN_2)A^{-1} \\ &= (AN_1A^{-1})(AN_2A^{-1}) \\ &= \phi_A(N_1)\phi_A(N_2). \end{aligned}$$

We can actually generalize the previous example to a type of automorphism.

Definition (Inner Automorphism Induced by a). Let G be a group and let $a \in G$. Then $\phi_a : G \to G$, defined by $\phi_a(x) = axa^{-1}$ for $x \in G$ is called the *inner automorphism of G induced by a*.

When G is a group, we use $\mathrm{Aut}(G)$ to denote the *set of all isomorphisms* of G and $\mathrm{Inn}(G)$ to denote the set of all *inner automorphisms* of G. The reason these sets are noteworthy is demonstrated by the next theorem.

Theorem 6.4 (Aut(G) and Inn(G) are Groups). The set of automorphisms of a group and the set of inner automorphisms of a group are both groups under the operation of function composition.

Finding $\mathrm{Inn}(G)$ is usually quite easy. If $G = \{e, a, b, c, \dots\}$, then $\mathrm{Inn}(G) = \{\phi_e, \phi_a, \phi_b, \phi_c, \dots\}$. However, finding $\mathrm{Aut}(G)$ can be quite involved.

Example 36. Let $G = (\mathbb{Z}_{10}, +)$. We have that

$$\mathrm{Inn}(G) = \{\phi_a : \mathbb{Z}_{10} \to \mathbb{Z}_{10} \mid \phi_a(x) = axa^{-1}, x \in \mathbb{Z}_{10}\}.$$

We have that for any $a \in \mathbb{Z}_{10}$

$$\phi_a(x) = a + x - a = x,$$

therefore $\phi_a(x) = x$ and

$$\mathrm{Inn}(\mathbb{Z}_{10}) = \{i_{\mathbb{Z}_{10}}\} = \{\mathrm{id}\}.$$

In general, if G is Abelian, then $\mathrm{Inn}(G)$ is just the identity mapping on the set G. What about $\mathrm{Aut}(G)$? Let $\phi : \mathbb{Z}_{10} \to \mathbb{Z}_{10}$ be an automorphism of $\mathbb{Z}_{10}$. We know that $\mathbb{Z}_{10}$ is generated by

elements with $\gcd(k,10)=1$, i.e.,

$$\begin{aligned}\mathbb{Z}_{10} &= \langle\overline{1}\rangle\\ &= \langle\overline{3}\rangle\\ &= \langle\overline{7}\rangle\\ &= \langle\overline{9}\rangle.\end{aligned}$$

Suppose that $\phi(\overline{1})=\overline{k}$. Since $G=\mathbb{Z}_{10}=\langle\overline{1}\rangle$ and ϕ is an isomorphism, $\langle\phi(\overline{1})\rangle=\overline{G}=G=\mathbb{Z}_{10}$. Let $\overline{m}\in\overline{G}=\mathbb{Z}_{10}$. We have that for all $\overline{m}\in\mathbb{Z}_{10}$

$$\begin{aligned}\phi(\overline{m}) &= \phi(\overline{1}+\overline{1}+\cdots+\overline{1})\\ &= \phi(\overline{1})+\phi(\overline{1})+\cdots+\phi(\overline{1})\\ &= m\cdot\overline{k}.\end{aligned}$$

Thus, ϕ depends on the value of $\overline{k}$. Since $\phi(\overline{1})=\overline{k}$, and $\overline{1}$ generates $\mathbb{Z}_{10}$, $\overline{k}$ must also generate $\mathbb{Z}_{10}$. Thus, $\overline{k}\in\{\overline{1},\overline{3},\overline{7},\overline{9}\}$. Therefore

$$\mathrm{Aut}(\mathbb{Z}_{10})=\{T_k:\mathbb{Z}_{10}\to\mathbb{Z}_{10}\mid T_k(\overline{a})=k\overline{a},\quad \gcd(k,10)=1,\overline{a}\in\mathbb{Z}_{10}\}$$

Essentially, we have that $\mathrm{Aut}(\mathbb{Z}_{10})$ is $\{1,3,7,9\}$, which is... $U(10)$!

As we saw in the previous example, the set of all automorphisms of $\mathbb{Z}_{10}$ appears to be the same as the set $U(10)$. This is actually true in general.

Theorem 6.5. For every positive integer n, $\mathrm{Aut}(\mathbb{Z}_n)$ is isomorphic to $U(n)$.

Proof. From the example above, any automorphism $\phi\in\mathrm{Aut}(\mathbb{Z}_n)$ is in the form $\phi(\overline{m})=m\cdot\overline{k}$, where $\overline{k}$ is $\phi(\overline{1})$ (or ϕ applied to any other generator of $\mathbb{Z}_n$). Let's define the isomorphism $\psi:\mathrm{Aut}(\mathbb{Z}_n)\to U(n)$ by $\psi(\phi)=\overline{k}$ where $\phi(\overline{1})=\overline{k}$. To show that ϕ is onto, let $\overline{U}(n)$. Thus, $\gcd(k,n)=1$. Choose $\phi=T_k$ where $T_k(\overline{x})=k\overline{x}$. Therefore, $\phi(\overline{1})=k\cdot\overline{1}=\overline{k}$ and we have $\psi(\phi)=\overline{k}\in U(n)$. □

Chapter 7

Cosets and Lagrange's Theorem

7.1 Properties of Cosets

In this chapter, we will prove the "single most important theorem in finite group theory" – Lagrange's Theorem. First, we must introduce a new tool for analyzing a group: the notion of a *coset*. This concept was invented by Galois in 1830, although the term was coined by G. A. Miller in 1910.

Definition (Coset of H in G). Let G be a group and let H be a nonempty subset of G. For any $a \in G$, the set $\{ah \mid h \in H\}$ is denoted aH. Analogously, $Ha = \{ha \mid h \in H\}$ and $aHa^{-1} = \{aha^{-1} \mid h \in H\}$. When H is a subgroup of G, the set aH is called the *left coset of H in G containing a*, whereas Ha is called the *right coset of H in G containing a*. In this case, the element a is called the *coset representation of aH* (or Ha). We use $|aH|$ to denote the number of elements in the set aH, and $|Ha|$ to denote the number of elements in Ha.

Let's give some examples to show how to find cosets.

Example 37. Let $G = (\mathbb{Z}, +)$ and let $H = 3\mathbb{Z} = \{3k \mid k \in \mathbb{Z}\}$. It's not hard to show that $H \leq G$. Consider $a = 0$. Then

$$aH = \{0 + h \mid h \in H\} = H.$$

If $a = 1$, then we would have

$$aH = 1H = \{1 + 3k \mid k \in \mathbb{Z}\}.$$

For $a = 3$, things get a little more interesting, as the pattern so far repeats.

$$\begin{aligned} 3H &= \{3 + 3k \mid k \in \mathbb{Z}\} \\ &= \{3(k^*) \mid k^* = k + 1 \in \mathbb{Z}\}. \end{aligned}$$

We can conclude that the left cosets of $G = \mathbb{Z}$ are

$$0 + H = H, 1 + H, 2 + H.$$

Since $G = (\mathbb{Z}, +)$ is abelian, the left cosets are the same as the right cosets.

Although we haven't considered the algebraic operations on cosets, we actually have that

$$\{H, 1+H, 2+H\} \cong \mathbb{Z}_3.$$

Intuitively, we can somewhat see this due to their modulo/repeating nature.

Remark 7. Modulo works for cosets as well. We have that

$$\begin{aligned} a \equiv b (\bmod H) &\iff ab^{-1} \in H \\ &\iff Ha = Hb. \end{aligned}$$

Let's consider another example.

Example 38. Let $G = S_3$ and let $H = \{(1),(12)\}$. Let's list all the elements of G

$$G = \{(1),(12),(13),(23),(123),(132)\}.$$

Let's find all left cosets of H in G

1. For $a = (1)$, we have that
$$aH = \{(1)(1),(1)(12)\} = H.$$

2. For $a = (12)$, things are a little less trivial
$$\begin{aligned} aH &= \{(12)(1),(12)(12)\} \\ &= \{(12),(1)\} \\ &= H. \end{aligned}$$

3. For $a = (13)$, we have
$$\begin{aligned} aH &= \{(13)(1),(13)(12)\} \\ &= \{(13),(123)\}. \end{aligned}$$

4. For $a = (23)$
$$aH = \{(23)(1),(23)(12)\} = \{(23),(132)\}.$$

5. For $a = (123)$
$$aH = \{(123)(1),(123)(12)\} = \{(123),(13)\}.$$

6. Finally, for $a = (132)$
$$aH = \{(132)(1),(132)(12)\} = \{(132),(23)\}.$$

Note that we actually have three repeating forms, which form the left cosets of H in G

$$H, (13)H, (23)H.$$

Similarly, we can find the right cosets.

Note that in general, we have that the left and right cosets are not equivalent, unless we are working with an Abelian group. Let's explore some more properties of cosets.

Lemma 2 (Properties of Cosets). Let H be a subgroup of G, and let $a, b \in G$. Then

1. $a \in aH$.
2. $aH = H$ if and only if $a \in H$.
3. $(ab)H = a(bH)$ and $H(ab) = (Ha)b$.
4. $aH = bH$ if and only if $a \in bH$.
5. $aH = bH$ or $aH \cap bH = \emptyset$.
6. $aH = bH$ if and only if $a^{-1}b \in H$.
7. $|aH| = |bH|$.
8. $aH = Ha$ if and only if $H = aHa^{-1}$.
9. aH is a subgroup of G if and only if $a \in H$.

Proof (Textbook).

1. $a = ae \in aH$.
2. Suppose $aH = H$. Then $a = ae \in aH = H$. Next, we want to show that if $a \in H$, $aH \subseteq H$ and $H \subseteq aH$. The former follows from closure of H. To show the latter, let $h \in H$. Since $a \in H$ and $h \in H$, we know that $a^{-1}h \in H$. Thus, $h = eh = (aa^{-1})h = a(a^{-1}h) \in aH$.
3. Follows directly from $(ab)h = a(bh)$ and $h(ab) = (ha)b$.
4. If $aH = bH$, then $a = ae \in aH = bH$. Conversely, if $a \in bH$, we have $a = bh$, where $h \in H$, and therefore $aH = (bh)H = b(hH) = bH$.
5. Follows directly from property 4, for if there is an element c in $aH \cap bH$, then $cH = aH$ and $cH = bH$.
6. $aH = bH$ if and only if $H = a^{-1}bH$. Then, the result follows from property 2.
7. We can just define a one-to-one mapping from aH onto bH. The correspondence $ah \to bh$ maps aH onto bH, and the fact that they are one-to-one follows directly from the cancellation property.
8. $aH = Ha$ if and only if $(aH)a^{-1} = (Ha)a^{-1} = H(aa^{-1} = H$ – that is, if and only if $aHa^{-1} = H$.
9. If aH is a subgroup, then it contains the identity element e. Thus, $aH \cap eH \neq \emptyset$, and by property 5 we have that $aH = eH = H$. Thus, from property 2, we have $a \in H$. Conversely, if $a \in H$, again by property 2, $aH = H$.

□

Proof (In Class).

5. Assume to the contrary that $aH \cap bH \neq \emptyset$. Then, there is $y \in aH \cap bH$. Suppose that $y = ah_1$ and $y = bh_2$ for some $h_1, h_2 \in H$. Thus, $ah_1 = y \in bH$. We have that $(ah_1)h_1^{-1} \in bHh^{-1} = bH$, i.e., $a \in bH$. From property 4, we have that $aH = bH$.

□

Note that properties 1, 5, and 7 of the lemma guarantee that the left cosets of a subgroup H of G

partition G into blocks of equal size. We may view the cosets of H as a partitioning of G into equivalence classes under the equivalence relation defined by $a \sim b$ if $aH = bH$.

The number of left (or right) cosets of a subgroup H in a group G is referred to as the *index* of a H in G. This number is denoted by $|G : H|$.

7.2 Lagrange's Theorem

Now, we can prove a theorem that has been around for longer than group theory itself.

Theorem 7.1 (Lagrange's Theorem). If G is a finite group and H is a subgroup of G, then $|H|$ divides $|G|$. Moreover, the number of distinct left/right cosets of H in G is $|G|/|H|$.

Proof. Since G is finite and $H \leq G$, there are a distinct finite number of left/right cosets: $H, a_1H, a_2H, \ldots, a_kH$ for some $k \in \mathbb{N}$ and $a_1, a_2, \ldots, a_k \in G$. We have $G = H \cup a_1H \cup \cdots \cup a_kH$, since $\{H, a_1H, a_2H, \ldots, a_kH\}$ is a complete list of left cosets. From the previous theorem, $|aH| = |bH|$ for all $a, b \in G$. Since $\{H, a_1H, a_2H, \ldots, a_kH\}$ is the list of disjoint cosets,

$$|G| = |H| + |a_1H| + |a_2H| + \cdots + |a_kH| = (k+1)|H|.$$

Thus, $|H| \mid |H|$.. □

Note however, that Lagrange's Theorem is a subgroup candidate criterion, i.e., it provides a list of candidates for the orders of the subgroups of a group. Thus, a group of order 12 may have subgroups of order 12, 6, 4, 3, 2, 1, but no others. Also note that the converse of Lagrange's Theorem is false. For example, a group of order 12 does not have to have a subgroup of order 6. Next, let's explore some corollaries of Lagrange's Theorem.

Corollary 10. If G is a finite group and H is a subgroup of G, then $|G : H| = |G|/|H|$.

Corollary 11. In a finite group, the order of each element of the group divides the order of the group.

Proof. Recall that the order of an element is the order of the subgroup generated by that element. □

Corollary 12. A group of prime order is cyclic.

Proof. Suppose that G has prime order. Let $a \in G$ and $a \neq e$, then $|\langle a \rangle|$ divides $|G|$ and $|\langle a \rangle| \neq 1$. Thus, $|\langle a \rangle| = |G|$ and the corollary follows. □

Corollary 13. Let G be a finite group and let $a \in G$. Then $a^{|G|} = e$.

Proof. By Corollary 2, $|G| = |a|k$ for some $k \in \mathbb{Z}^+$. Thus, $a^{|G|} = a^{|a|k} = e^k = e$. □

Another corollary of Lagrange's Theorem is *Fermat's Little Theorem*, which has been used in conjunction with computers to test for primality of certain numbers

Corollary 14 (Fermat's Little Theorem). For every integer a and every prime p, $a^p \bmod p = a \bmod p$.

Proof. Let p be a prime number and $a \in \mathbb{Z}$. If $p \mid a$, we have that

$$a^p \equiv 0 \equiv a \pmod{p}.$$

In the case that $p \nmid a$, consider $U(p) = \{1, 2, \ldots, p-1\}$. We have that $|U(p)| = p - 1$. By the previously corollary that $a^{|}G| = e$, we have that

$$\begin{aligned} a^{p-1} &\equiv \overline{1} \pmod{p} \\ a \cdot a^{p-1} &\equiv a \cdot \overline{1} \pmod{p} \\ a^p &\equiv a \pmod{p} \end{aligned}$$

□

Langrage's Theorem demonstrates that the finiteness of a group imposes severe restrictions on the possible orders of subgroups. The next theorem places limits on the existence of certain subgroups in finite groups.

Theorem 7.2. For subgroups, H and K of a finite group, define the set $HK = \{hk \mid h \in H, k \in K\}$. Then

$$|HK| = \frac{|H||K|}{|H \cap K|}.$$

Proof. Define $f : H \times K \to HK$ by $f(h,k) = hk$ for any $h \in H$ and $k \in K$, where we define $H \times K = \{(h,k) \mid h \in H, k \in K\}$. Let $|H \cap K| = n \in \mathbb{N}$ and $H \cap K = \{t_1, t_2, \ldots, t_n\}$. We will show that f is a n-to-one map onto HK, i.e., $\left|f^{-1}(\{x\})\right| = n$ for each $x \in HK$. Let $x = hk \in HK$ where $h \in H$ and $k \in K$. There are n different ordered pairs:

$$(ht_1, t_1^{-1}k), (ht_2, t_2^{-1}k), \ldots, (ht_n, t_n^{-1}k)$$

such that

$$f((ht_i, t_i^{-1}k)) = (ht_i)(t_i^{-1}k) = kh, \quad \forall i \in \{1, 2, \ldots, n\}.$$

Thus, f is an n-to-one map from $H \times K$ onto HK. Thus, we have that

$$|HK| = \frac{|H \times K|}{|H \cap H|} = \frac{|H||K|}{|H \cap K|}.$$

□

Let's see how we can apply this theorem with an example.

Example 39. Let's consider a group G with $|G| = 75$. Can G have more than one subgroup of order 25? The answer is *no*. Why? Assume to the contrary that there are $H, K \leq G$ and $|H| = |K| = 25$. Since $|H \cap K|$ divides $|H| = 25$. We know from Lagrange's Theorem that

$|H \cap K| = 1$ or 5, so this results in

$$\frac{|H||K|}{|H \cap K|} = \frac{25 \cdot 25}{|H \cap K|} = 625 \text{ or } 125 \text{ elements.}$$

Therefore, we have that $|H \cap K| = 25$ and therefore, $H = K$.

For any prime $p > 2$, we know that Z_{2p} and D_p are nonisomorphic groups of order $2p$. This naturally raises the question of whether there could be other possible groups of these orders. With the tools available to us at this point, we can answer this question.

Theorem 7.3 (Classification of Groups of Order $2p$). Let G be a group of order $2p$, where p is a prime greater than 2. Then G is isomorphic to Z_{2p} or D_p.

Proof. If G has an element g with $|g| = 2p$, then $G = \langle g \rangle$. We are done, as $G \cong Z_{2p}$. If G is not cyclic, then we would like to show that $G \cong D_p$. We will show that G has an element g with $|g| = p$. Assume to the contrary that any element of G has order not equal to p. By Lagrange's Theorem, for any element $g \in G$, $|g| \mid 2p$. Since $|g| \neq p$, $|g| = 1$ or 2. Thus, $\forall g \in G \setminus \{e\}$, $|g| = 2$. This implies that G is Abelian[a]. Choose non-identity elements a, b such that $a \neq b$. Since a, b, and ab have order 2 and G is Abelian, we have that $\{e, a, b, ab\}$ is a subgroup of G. Since $H \leq G$ and $|H| = 4$, by Langrage's Theorem, $|H| \big| |G| = 2p$. This is a contradiction. Thus, there is $g \in G$ such that $|g| = p$. Choose $a \in G$ such that $a \notin \langle g \rangle$. We have that $|a| = 2, p$. If $|a| = p$, then we also have $\langle a \rangle \cap \langle g \rangle = \{e\}$. We have that

$$\begin{aligned} |\langle a \rangle \langle g \rangle| &= \frac{|\langle a \rangle| |\langle g \rangle|}{|\langle a \rangle \cap \langle g \rangle|} \\ &= \frac{p \cdot p}{1} \\ &= p^2, \end{aligned}$$

which is a contradiction. Thus, $|a| = 2$. We want to show that $ga = ag^{-1}$. Observe that $G = \langle a, g \rangle$ because $\langle a \rangle, \langle g \rangle \leq G$, we have that

$$\begin{aligned} |\langle a \rangle \langle g \rangle| &= \frac{|\langle a \rangle| |\langle g \rangle|}{|\langle a \rangle \cap \langle g \rangle|} \\ &= \frac{2 \cdot p}{1} \\ &= 2p. \end{aligned}$$

Thus, we have $2p \leq |\langle a, g \rangle| \leq |G| = 2p$. Thus, $G = \langle a, g \rangle$. Now, showing $ga = ag^{-1}$ is relatively simple

$$\begin{aligned} (ga)^{-1} &= a^{-1} g^{-1} \\ &= ag^{-1} && (|a| = 2) \end{aligned}$$

Note that $ga \notin \langle g \rangle$, as if it were, then $a = g^{-1} ga \in \langle g \rangle$, which is a contradiction since we chose $a \in G$ such that $a \notin \langle g \rangle$. Thus, $|ga| = 2$. Thus, $(ga)^{-1} = ga = ag^{-1}$. Therefore, G and D_p have the same group structure where g acts as a rotation and a acts as a reflection. We have $G \cong D_p$. □

[a] $ab = (ab)^{-1} = b^{-1} a^{-1} = ba$

We will now touch upon another related theorem which is meant to be introduced in Chapter 24, however, it is directly related to Theorem 7.3.

7.3 The Sylow Theorems

First, let's introduce the relevant terminology.

Definition (p-group). Let G be a group and let p be a prime. A group of order p^k for some $k \geq 1$ is called a *p-group*. Similarly, a subgroup of order p^j for some $j \geq 1$ is called a *p-subgroup*. If $|G| = p^\alpha m$ where $\gcd(p, m) = 1$, then a subgroup of order p^α is called a *Sylow p-subgroup of G*.

In terms of notation, we denote the set of all Sylow p-subgroups of a group G as $\mathrm{Syl}_p(G)$. The size of this set, $|\mathrm{Syl}_p(G)$, is denoted $n_p(G)$. With these concepts in mind, we can introduce the Sylow theorems, which is one of the most important results in finite group theory, alongside Lagrange's Theorem. We begin with Sylow's First Theorem (1872), which shows that Sylow p-subgroups exist.

Theorem 24.3 (Existence of Subgroups of Prime-Power-Order). Let G be a finite group and let p be a prime. if p^k divides $|G|$, then G has at least one subgroup of order p^k.

Next is Sylow's Second Theorem, which states that all Sylow p-subgroups are conjugate in G.

Theorem 24.4 (Sylow's Second Theorem). If H is a subgroup of a finite group G and $|H|$ is a power of a prime p, then H is contained in some Sylow p-subgroup of G.

Finally, Sylow's Third Theorem states that

Theorem 24.5 (Sylow's Third Theorem). Let p be a prime and let G be a group of order $p^k m$, where p does not divide m. Then the number n of Sylow p-subgroups of G is equal to 1 modulo p and divides m. Furthermore, any two Sylow p-subgroups of G are conjugate.

These theorems have accelerated the classification of finite groups as they have helped us to find many more subgroups.

Chapter 8

External Direct Products

8.1 Definition and Examples

In this chapter, we will show how to piece together groups to make larger groups. In chapter 9, we will show that we can often start with one large group and decompose it into the product of smaller groups (much in the same way that a composite positive integer can be broken down into a product of primes). These methods will later be used to give us a simple way to construct all finite Abelian groups.

Definition (External Direct Product). Let $G_1, G_2, \ldots, G_n$ be a finite collection of groups. The *external direct product* of $G_1, G_2, \ldots, G_n$, written as $G_1 \oplus G_2 \oplus \cdots \oplus G_n$, is the set of all n-tuples for which the ith component is an element of G_i and the operation is componentwise.

In symbols,

$$G_1 \oplus G_2 \oplus \cdots \oplus G_n = \{(g_1, g_2, \ldots, g_n) \mid g_i \in G_i\},$$

where $(g_1, g_2, \ldots, g_n)(g_1', g_2', \ldots, g_n')$ is defined to be $(g_1 g_1', g_2 g_2', \ldots, g_n g_n')$. Each product $g_i g_i'$ uses the operation of G.

Example 40. As an example, $\mathbb{R}^2 = \mathbb{R} \oplus \mathbb{R}$ using addition as the operation. We can construct $\mathbb{C}$ with this as well

$$\begin{aligned} \mathbb{C} &= \{x + iy \mid x, y \in \mathbb{R}\} \\ &\cong \{(x, y) \mid x, y \in \mathbb{R}\}. \end{aligned}$$

Example 41. Consider $\mathbb{Z}_3 \oplus \mathbb{Z}_2 = \{(0,0), (0,1), (1,0), (1,1), (2,0), (2,1)\}$. We can perform operations on these elements, e.g.,

$$(0,1) + (2,1) = (2,2) = (0,0).$$

Example 42. Consider $U(3) \oplus U(4)$, which is

$$\{1,2\} \oplus \{1,3\} = \{(1,1), (1,3), (2,1), (2,3)\}.$$

We have that

$$\begin{aligned}(1,3)\oplus(2,1) &= (1\times 2, 3\times 1)\\ &= (2,3).\end{aligned}$$

Example 43. Let G be a group with $|G| = 4$. We have that either $G \cong \mathbb{Z}_4$ or $G \cong \mathbb{Z}_2 \oplus \mathbb{Z}_2$.

8.2 Properties of External Direct Products

We begin with a theorem that gives us a method to compute the order of an element in a direct product.

Theorem 8.1 (Order of An Element in a Direct Product). The order of an element in a direct product of a finite number of finite groups is the least common multiple of the orders of the components of the element. In symbols,

$$|(g_1, g_2, \ldots, g_n)| = \operatorname{lcm}(|g_1|, |g_2|, \ldots, |g_n|).$$

Proof. Let $s = \operatorname{lcm}(|g_1|, |g_2|, \ldots, |g_n|)$ and $t = |(g_1, g_2, \ldots, g_n)|$. Observe that

$$\begin{aligned}(g_1, g_2, \ldots, g_n)^s &= (g_1^s, g_2^s, \ldots, g_n^s)\\ &= (e_1, e_2, \ldots, e_n) && (|g_i| \mid s)\end{aligned}$$

Thus, $t \le s$. On the other hand

$$\begin{aligned}(g_1^t, g_2^t, \ldots, g_n^t) &= (g_1, g_2, \ldots, g_n)^t\\ &= (e_1, e_2, \ldots, e_n).\end{aligned}$$

Looking at this component wise, we have that $g_i^t = e_i$, for all $i = 1, 2, \ldots, n$. We have that $|g_i| \mid t$ for all $i = 1, 2, \ldots, n$. Therefore, t is a common multiple of $|g_i|$'s and we have that $s \le t$ since $s = \operatorname{lcm}(|g_1|, \ldots, |g_n|)$ by definition. Therefore, we have that $s = t$. □

This is a very useful result. Let's take a look at an example.

Example 44. Consider some groups of order 100

$$\mathbb{Z}_{100}, \quad \mathbb{Z}_{25}\oplus\mathbb{Z}_4, \quad \mathbb{Z}_5\oplus\mathbb{Z}_5\oplus\mathbb{Z}_4, \quad \mathbb{Z}_5\oplus\mathbb{Z}_{20}.$$

Is $\mathbb{Z}_{25}\oplus\mathbb{Z}_4$ isomorphic to $\mathbb{Z}_{20}\oplus\mathbb{Z}_5$? It turns out that the answer is *no*. Observe that 25 and 4 are relatively prime, but 20 and 5 are not. For $\mathbb{Z}_{25}\oplus\mathbb{Z}_4$, consider $(1,1) \in \mathbb{Z}_{25}\oplus\mathbb{Z}_4$. For $\mathbb{Z}_{20}\oplus\mathbb{Z}_5$ consider the general element (g_1, g_2). The order of $(1,1)$ is

$$\begin{aligned}|(1,1)| &= \operatorname{lcm}(|1|_{\mathbb{Z}_{25}}, |1|_{\mathbb{Z}_4})\\ &= \operatorname{lcm}(25, 4)\\ &= 100.\end{aligned}$$

For (g_1, g_2)

$$\begin{aligned}|(g_1, g_2)| &= \operatorname{lcm}(|g_1|, |g_2|)\\ &\leq \operatorname{lcm}(20, 5)\\ &= 20.\end{aligned}$$

Therefore, no matter the element that we pick in $\mathbb{Z}_{20} \oplus \mathbb{Z}_5$, its order cannot exceed 20. Since isomorphisms preserve order, there is no way the two groups can be isomorphic.

The next theorem characterizes direct products of cyclic groups that are themselves cyclic.

Theorem 8.2 (Criterion for $G \oplus H$ to be Cyclic). Let G and H be finite cyclic groups. Then $G \oplus H$ is cyclic if and only if $|G|$ and $|H|$ are relatively prime.

Proof. Let $|G| = m$ and $|H| = n$. We have that $|G \oplus H| = mn$.

($\Rightarrow$)

Assume that $G \oplus H$ is cyclic. Thus, there is $(g, h) \in G \oplus H$ such that $|(g, h)| = mn$ (i.e., (g, h) is a generator). We want to show that m and and n are relatively prime. Let $d = \gcd(m, n)$. Note that $\frac{mn}{d} \in \mathbb{N}$. Thus

$$\begin{aligned}(g, h)^{\frac{mn}{d}} &= ((g^m)^{\frac{n}{d}}, (h^n)^{\frac{m}{d}})\\ &= (e_1^{\frac{n}{d}}, e_2^{\frac{m}{d}})\\ &= (e_1, e_2).\end{aligned}$$

Thus, we have that $|(g, h)| = mn$ divides $\frac{mn}{d}$. Therefore, it must be the case that $d = 1$[a].

($\Leftarrow$)

Assume that m and n are relatively prime. Let $G = \langle g \rangle$ and $H = \langle h \rangle$. From Theorem 8.1

$$\begin{aligned}|(g, h)| &= \operatorname{lcm}(|g|, |h|)\\ &= \operatorname{lcm}(m, n)\\ &= mn. && (m, n \text{ relatively prime})\end{aligned}$$

Thus, $G \oplus H = \langle (g, h) \rangle$ since $|(g, h)| = |G \oplus H|$. □

[a] If $d > 1$, then $\frac{mn}{d}$ would be smaller than mn, so mn couldn't divide it.

A consequence of this theorem (combined with induction) gives us the following corollary.

Corollary 15 (Criterion for $G_1 \oplus G_2 \oplus \cdots \oplus G_n$ to be Cyclic). An external direct product $G_1 \oplus G_2 \oplus \cdots \oplus G_n$ of a finite number of finite cyclic groups is cyclic if and only if $|G_i|$ and $|G_j|$ are relatively prime when $i \neq j$.

We also have another corollary which is essentially just a rewrite of the previous for the case of $\mathbb{Z}_n$.

Corollary 16 (Criterion for $\mathbb{Z}_{n_1 n_2 \ldots n_k} \cong \mathbb{Z}_{n_1} \oplus \mathbb{Z}_{n_2} \oplus \cdots \oplus \mathbb{Z}_{n_k}$). Let $m = n_1 n_2 \ldots n_k$. Then

$\mathbb{Z}_m$ is isomorphic to $\mathbb{Z}_{n_1} \oplus \mathbb{Z}_{n_2} \oplus \cdots \oplus \mathbb{Z}_{n_k}$ if and only if n_i and n_j are relatively prime when $i \neq j$.

Let's do an example.

Example 45. We know that $\mathbb{Z}_{100} \cong \mathbb{Z}_{25} \oplus \mathbb{Z}_4$ since $100 = 25 \cdot 4$ and $\gcd(25,4) = 1$. On the other hand, $\mathbb{Z}_{100} \ncong \mathbb{Z}_{20} \oplus \mathbb{Z}_5$ since although $20 \cdot 5 = 100$, we have that $\gcd(20,5) \neq 1$.

8.3 The Group of Units Modulo n as External Direct Products

The U-groups provide a convenient way to illustrate the ideas we've just introduced. First, we introduce some new notation. Suppose $k \mid n$, then

$$U_k(n) = \{x \in U(n) \mid x \bmod k = 1\}.$$

It can be shown that $U_k(n) \leq U(n)$.

Theorem 8.3 ($U(n)$ as an External Direct Product). Suppose s and t are relatively prime. Then $U(st) \cong U(s) \oplus U(t)$. Moreover, $U_s(st) \cong U(t)$ and $U_t(st) \cong U(s)$.

Proof. Let s and t be relatively prime. We can create an isomorphism from $U(st)$ to $U(s) \oplus U(t)$ to be $\phi : U(st) \to U(s) \oplus U(t)$ defined as

$$\phi(x) \to (x \bmod s, x \bmod t), \quad x \in U(st).$$

We want to show that ϕ is an isomorphism.

1. *One-to-one*: let $x, y \in U(st)$ such that $\phi(x) = \phi(y)$. We have that

$$\begin{aligned} \phi(x) &= \phi(y) \\ (x \bmod s, x \bmod t) &= (y \bmod s, y \bmod t). \end{aligned}$$

   Considering the equalities pairwise, we have

$$\begin{aligned} x \bmod s &= y \bmod s \\ x \bmod t &= y \bmod t. \end{aligned}$$

   This means that $s \mid x - y$ and $t \mid x - y$. Since s and t are relatively prime, by Euclid's Lemma[a] $st \mid x - y$. Thus

$$x - y \equiv 0 \pmod{st} \iff x \equiv y \pmod{st}.$$

   Thus, we have $x_1 = x_2$ in $U(st)$.

2. *Onto*: LAETR.

3. *Operation preserving*: consider $x, y \in U(st)$. We have that

$$\begin{aligned} \phi(xy) &= (xy \bmod s, xy \bmod t) \\ &= ((x \bmod s)(y \bmod s) \bmod s, (x \bmod t)(y \bmod t) \bmod t) \\ &= \phi(x)\phi(y) \end{aligned}$$

□

[a] See Chapter 0, probably.

As a consequence of Theorem 8.3, we have that

Corollary 17. Let $m = n_1 n_2 \dots n_k$, where $\gcd(n_i, n_j) = 1$ for $i \neq j$. Then

$$U(m) \cong U(n_1) \oplus U(n_2) \oplus \cdots \oplus U(n_k).$$

Chapter 9

Normal Subgroups and Factor Groups

9.1 Normal Subgroups

We saw in Chapter 7 that if G is a group and $H \leq G$, it is not always true that $aH = Ha$ for all $a \in G$. However, there are certain situations where this does hold. Consider $G = \mathbb{Z}$ and $H = \{5k \mid k \in \mathbb{Z}\}$. It has the cosets

$$\{0+H, 1+H, 2+H, 3+H, 4+H\} = \{\overline{0}, \overline{1}, \overline{2}, \overline{3}, \overline{4}\}.$$

When dealing with any integer related problem, we generally handle things using modulo. Can we do the same for groups? It turns out that *yes*, if we have a *normal subgroup*. 185 years ago, Galois first recognized that such subgroups were worthy of special attention.

Definition (Normal Subgroup). A subgroup H of a group G is called a *normal subgroup* of G if $aH = Ha$ for all a in G. We denote this by $H \lhd G$.

Example 46. Every subgroup of an Abelian group is normal. In this case, if G Abelian and $H \leq G$, then $ah = ha$ for $a \in G$ and $h \in H$.

Example 47. Consider $G = S_5$ and $H = A_5$ where $H \leq G$. Is $A_5 \lhd S_5$? We want to show that $\alpha A_5 = A_5 \alpha$ for all $\alpha \in S_5$. It suffices to show that $\alpha A_5 \alpha^{-1} = A_5$ for all $\alpha \in S_5$. Let $\alpha \in S_5$ and $\beta \in A_5$ We want to show that $\alpha\beta\alpha^{-1} \in A_5$. We know that β is even and that α/α^{-1} have the same parity. Therefore, $\alpha\beta\alpha^{-1}$ is even, and hence in A_5. For the converse, we want to show that $\beta \in A_5$ implies $\beta \in \alpha A_5 \alpha^{-1}$ for $\alpha \in S_5$. We have that

$$\beta = \alpha(\alpha\beta\alpha)\alpha^{-1} \in \alpha A_5 \alpha^{-1}$$

We can conclude that $\alpha A_5 \alpha^{-1} = A_5$ for $\alpha \in S_5$, therefore $A_5 \lhd S_5$.

This is actually the test for determining normal subgroups.

Theorem 9.1 (Normal Subgroup Test). $H \lhd G$ if and only if $xHx^{-1} \subseteq H$ for all $x \in G$.

Proof.

($\Rightarrow$)

Since $H \lhd G$, we have that $xH = Hx$ for all $x \in G$. Hence, $xHx^{-1} = H$ for all $x \in G$. Therefore, $xHx^{-1} \subseteq H$ for all $x \in G$.

($\Leftarrow$)

Assume that $xHx^{-1} \subseteq H$ for all $x \in G$. Let $x = a \in G$. Then $aHa^{-1} \subseteq H$, i.e., $aH \subseteq Ha$. Similarly, let $x \in a^{-1} \in G$. Then $a^{-1}Ha \subseteq H$, i.e., $Ha \subseteq aH$. □

Before continuing, we have a quick remark on notation.

Remark 8. The following are equivalent

1. $H \lhd G$
2. $\forall x \in G, xH = Hx$
3. $\forall x \in G, xHx^{-1} = H$
4. $\forall x \in G, xHx^{-1} \subseteq H$

Let's look at a couple more examples.

Example 48. Recall the center of a group G

$$Z(G) = \{z \in G \mid \forall g \in G, gz = za\}.$$

Is $Z(G) \lhd G$? Yes! We want to show that $xZ(G)x^{-1} \subseteq Z(G)$ for $x \in G$. Let $x \in G$ and $z \in Z(G)$. Then

$$xzx^{-1} = x^{-1}xz = z \in Z(G).$$

Therefore, $xZ(G)x^{-1} \subseteq Z(G)$ and we have $Z(G) \lhd G$.

Example 49. $A_n \lhd S_n$ for any $n \in \mathbb{N}$.

Example 50. Let $A = \{\text{rotations in } D_n\}$. Clearly $A \leq D_n$. For any rotation $R \in A$ and reflection F, we have $FR = R^{-1}F$. Thus, $FRF^{-1} = R^{-1} \in A$. We can conclude that $xAx^{-1} \subseteq A$ for all $x \in D_n$. Thus, $A \lhd D_n$.

9.2 Factor Groups

Why are normal subgroups of special significance? When the subgroup H of G is normal, then the set of left (or right) cosets of H in G is itself a group – called the *factor group of G by H* (or the *quotient group of G by H*. One can often obtain information about a group by studying one of its factor groups.

Theorem 9.2 (Factor Groups (Holder, 1889)). Let G be a group and let H be a normal subgroup of G. The set $G/H = \{aH \mid a \in G\}$ is a group under the operation $(aH)(bH) = abH$.

Proof. We want to show that G/H is well defined. Let $a, a', b, b' \in G$. Assume that $aH = a'H$ and $bH = b'H$. We want to show that $abH = a'b'H$. Since $aH = a'H$ and $bH = b'H$, there are $h_1, h_2 \in H$ such that $a' = ah_1$ and $b' = bh_2$.

$$\begin{aligned} a'b'H &= (ah_1)(bh_2)H \\ &= ah_1bH \\ &= ah_1(Hb) \\ &= a(h_1H)b \\ &= a(Hb) \\ &= abH. \end{aligned}$$

Thus, the operation is well defined. We will show that G/H is a group.

1. *Identity*: we have $eH = H$ as the identity of G/H.
2. *Associativity*: let $aH, bH, cH \in G/H$.

$$\begin{aligned} (aHbH)cH &= (abH)cH \\ &= (ab)cH \\ &= a(bc)H \\ &= a(bH)cH \\ &= aH(bHcH) \end{aligned}$$

3. *Inverse*: we claim that $a^{-1}H$ is an inverse.

$$\begin{aligned} (aH)(a^{-1}H) &= aa^{-1}H \\ &= eH \\ &= H. \end{aligned}$$

   Similarly, $(a^{-1}H)(aH) = H$.

Thus $G/H = \{aH \mid a \in G\}$. □

When we factor out by a normal subgroup H, what we are essentially doing is defining every element in H to be the *identity*. Algebraists often refer to the process of creating the factor G/H as *killing* H.

9.3 Applications of Factor Groups

One consequence of the fact that the operation for a factor group G/H is inherited G is that many properties of G are inherited by G/H and many properties of G can be deduced from properties of G/H. This is good as the structure of G/H is usually less complicated than that of G, yet it serves as an approximation for G in many ways*.

*Perhaps in the way 3.14 is an approximation of π

Theorem 9.3 (G/Z Theorem). Let G be a group and let $Z(G)$ be a center of G. If $G/Z(G)$ is cyclic, then G is Abelian.

Proof. Note that G abelian if and only if $Z(G) = G$. Since $G/Z(G)$ is cyclic, there is $g \in G$ such that $G/Z(G) = \langle gZ(G) \rangle$. Let $a \in G$. We have that there is $i \in \mathbb{Z}$ such that

$$\begin{aligned} aZ(G) &= (gZ(G))^i \\ &= g^i Z(G). \end{aligned}$$

Thus, $a = g^i z$ for some $z \in Z(G)$. Similarly, any $b \in G$, $b = g^j z'$ for some $j \in \mathbb{Z}$ and $z' \in Z(G)$. We have

$$\begin{aligned} ab &= (g^i z)(g^j z') \\ &= g^i g^j z z' \\ &= g^j g^i z' z \\ &= (g^j z')(g^i z) \\ &= ba \end{aligned}$$

□

Theorem 9.4 (G/Z(G) ≅ Inn(G)). For any group G, $G/Z(G)$ is isomorphic to $\mathrm{Inn}(G)$.

Proof. Let $g \in G$. Define $T : G/Z(G) \to \mathrm{Inn}(G)$ by $T(gZ(G)) = \phi_g$ where $\phi_g(x) = gxg^{-1}$ for all $x \in G$. We will show that T is an isomorphism.

1) *T is well defined*: assume that $gZ(G) = hZ(G)$. We want to show that $\phi_g = \phi_h$. Note that $h^{-1}g \in Z(G)$. For all $x \in G$, we have that

$$\begin{aligned} (h^{-1}g)x &= x(h^{-1}g) \\ h(h^{-1}gx)g^{-1} &= h(xh^{-1}g)g^{-1} \\ gxg^{-1} &= hxh^{-1} \\ \phi_g(x) &= \phi_h(x). \end{aligned}$$

   Thus, $\phi_g = \phi_h$.

2) *T bijective*: assume that $\phi_{g_1} = \phi_{g_2}$. We want to show that $g_1 Z(G) = g_2 Z(G)$. This is equivalent to showing that $g_2^{-1} g_1 \in Z(G)$. For all $x \in G$, we have

$$\begin{aligned} \phi_{g_1}(x) &= \phi_{g_2}(x) \\ g_1 x g_1^{-1} &= g_2 x g_2^{-1} \\ g_2^{-1}(g_1 x g_1^{-1}) g_1 &= g_2^{-1}(g_2 x g_2^{-1}) g_1 \\ g_2^{-1} g_1 x &= x g_2^{-1} g_1. \end{aligned}$$

   Thus, $(g_2^{-1} g_1)x = x(g_2^{-1} g_1)$ for all $x \in G$. We have $g_2^{-1} g_1 \in Z(G)$. Thus $g_1 Z(G) = g_2 Z(G)$.

3) *T is onto*: let $\phi_g \in \mathrm{Inn}(G)$. Choose $gZ(G) \in G/Z(G)$. Then $T(gZ(G)) = \phi_g$.

4) *T is operation preserving*:

$$\begin{aligned} T(gZ(G)h(Z(G))(x) &= T(ghZ(G))(x) \\ &= \phi_{gh}(x) \\ &= (gh)x(gh)^{-1} \\ &= g(hxh^{-1})g^{-1} \\ &= g\phi_h(x)g^{-1} \\ &= \phi_g(\phi_n(x)) \\ &= (\phi_g \circ \phi_h)(x) \end{aligned}$$

□

The next theorem demonstrates one of the most powerful proof techniques available in the theory of finite groups – the combined use of factor groups and induction.

Theorem 9.5 (Cauchy's Theorem for Abelian Groups). Let G be a finite Abelian group and let p be a prime that divides the order of G. Then G has an element of order p.

Proof Sketch. Choose $x \in G \setminus \{e\}$. Let $H = \langle x \rangle$. Consider G/H. By mathematical induction, there is $\overline{H} \leq G/H$ with $|\overline{H}| = p$. □

Cauchy's Theorem is nice because it guarantees a subgroup of prime order p, if p divides the order of the group. Note that it is a special case of Sylow's Theorem: p^i divides G implies that $\exists H \leq G, |H| = p^i$.

9.4 Internal Direct Products

We've shown that we can use external direct products as a method to put groups together to get a larger group, in such a way that we can determine many properties of the larger group from the properties of its components. What there was a way to reverse this – to start with a large group G and break it down into a product of subgroups? It is occasionally possible to do that.

Definition (Internal Direct Product of H and K). We say that G is the internal direct product of H and K and write $G = H \times K$ if H and K are normal subgroups of G and

$$G = HK \quad \text{and} \quad H \cap K = \{e\}.$$

How does this differ from an *external* direct product? Well, to form an interal direct product, we start with a group G and then proceed to find two subgroups H and K within G such that G is *isomorphic* to the external direct product of H. On the other hand, to form an external direct product, we start with any two groups H and K (related or not) and produce the larger group $H \oplus K$.

A group G can also be the internal direct product of a collection of subgroups.

Definition (Internal Direct Product of $\mathbf{H_1 \times H_2 \times \cdots \times H_n}$). Let $H_1, H_2, \ldots, H_n$ be a finite collection of normal subgroups of G. We say that G is the *internal direct product* of

$H_1, H_2, \ldots, H_n$ and write $G = H_1 \times H_2 \times \cdots \times H_n$, if

1. $G = H_1 H_2 \ldots H_n = \{h_1 h_2 \ldots h_n \mid h_i \in H_i\}$
2. $(H_1 H_2 \ldots H_i) \cap H_{i+1} = \{e\}$ for $i = 1, 2, \ldots, n-1$

The reason that we want each subgroup to be normal and disjoint from the product of all previous ones is to keep the internal direct product isomorphic to the external direct product.

Theorem 9.6. If a group G is the internal direct product of a finite number of subgroups $H_1, H_2, \ldots, H_n$, then G is isomorphic to the external direct product of $H_1, H_2, \ldots, H_n$.

Proof ($n = 2$). Assume that $G = H \times K$, i.e., $H, K \lhd G$, $HK = G$ and $H \cap K = \{e\}$. Define $T : H \times K \to H \oplus K$ as follows: for each $hk \in H \times K$, $T(hk) = (h, k) \in H \oplus K$. We will show that T is an isomorphism.

1) *T is well-defined*: assume that $hk = h'k'$. We want to show that $(h, k) = (h', k')$. Since $hk = h'k'$, $(h')^{-1}h = k'k^{-1}$. Since the former is in H and the latter is in K, $(h')^{-1}h, k'k^{-1} \in H \cap K = \{e\}$. We have that $(h')^{-1}h = e = k'k^{-1}$.
2) *T one-to-one*: assume that $T(h_1k_1) = T(h_2k_2)$ where $h_1, h_2 \in H$ and $k_1, k_2 \in K$. Thus, $(h_1, k_1) = (h_2, k_2)$. Therefore, $h_1 = h_2$ and $k_1 = k_2$. We have $h_1k - 1 = h_2k_2$ as well.
3) *T is onto*: let $(h, k) \in H \oplus K$. Choose $hk \in H \times K = HK$. Then $T(hk) = (h, k)$.
4) *T is operation preserving*: consider $h_1k_1, h_2k_2 \in H \times K$. If $*$ is the operation in $H \times K$ and $\cdot$ is the operation in $H \oplus K$,

$$\begin{aligned}
(h_1k_1) * (h_2k_2) &= (h_1h_2)(k_1k_2) \\
T((h_1k_1) * (h_2k_2)) &= T((h_1h_2)(k_1k_2)) \\
&= (h_1h_2, k_1k_2) \\
&= (h_1, k_1) \cdot (h_2, k_2).
\end{aligned}$$

□

The next theorem provides an important application of Theorem 9.6.

Theorem 9.7 (Classification of Groups of Order $\mathbf{p^2}$). Every group of order p^2, where p is a prime, is isomorphic to $\mathbb{Z}_{p^2}$ or $\mathbb{Z}_p \oplus \mathbb{Z}_p$.

An immediate corollary of this is

Corollary 18. If G is a group of order p^2, where p is a prime, then G is Abelian.

Chapter 10

Group Homomorphisms

A homomorphism* is a natural generalization of an isomorphism. It is one of the most fundamental ideas of algebra. It was first introduced by Camille Jordan in 1870.

Definition (Group Homomorphism). A *homomorphism* ϕ from a group G to a group $\overline{G}$ is a mapping from G onto $\overline{G}$ that preserves the group operation; that is, $\phi(ab) = \phi(a)\phi(b)$ for all a, b in G.

Let's see some examples.

Example 51. The homomorphism $\phi : \mathbb{Z} \to \mathbb{Z}_5$ is defined by $\phi(x) = x \pmod 5$. This is obviously not bijective, so ϕ is not an isomorphism, but we have that

$$\phi(ab) = \phi(a)\phi(b), \quad \forall a, b \in \mathbb{Z}.$$

We should also remark that every isomorphism is a homomorphism.

Remark 9. Note that if $\phi : G \to \overline{G}$ is an isomorphism, then ϕ is a homomorphism.

Example 52. Let $\phi : \mathbb{R}[x] \to \mathbb{R}[x]$ defined by $\phi(f(x)) = f'(x)$, where

$$\mathbb{R}[x] = \{f : \mathbb{R} \to \mathbb{R} \mid \text{f is a polynomial}\}.$$

We have that for $f, g \in \mathbb{R}[x]$

$$\begin{aligned}\phi(f(x) + g(x)) &= [f(x) + g(x)]' \\ &= f'(x) + g'(x) \\ &= \phi(f(x)) + \phi(g(x)).\end{aligned}$$

So ϕ is a homomorphism. Is ϕ an isomorphism? No. We can show that ϕ is not injective. Consider $f(x) = x^2 + 2, g(x) = x^2 + 5 \in \mathbb{R}[x]$. We have that

$$\phi(f(x)) = 2x = \phi(g(x)),$$

*From the Greek *homo* (like) and *morphe* (form).

but $f(x) \neq g(x)$. Therefore, ϕ is not injective, and hence is not an isomorphism.

One important subgroup that is closely related to the image of a homomorphism is the *kernel* of a homomorphism.

Definition (Kernel of a Homomorphism). The *kernel* of a homomorphism ϕ from a group G to to a group with identity e is the set $\{x \in G \mid \phi(x) = e\}$. The kernel of ϕ is denoted by $\operatorname{Ker}\phi$.

Note that every linear transformation (in linear algebra) is a group homomorphism and the null-space is the same as the kernel. A transformation that is invertible is a group isomorphism.

10.1 Properties of Homomorphisms

First, we introduce a relevant definition.

Definition. Let $g \in G$. Define $g\operatorname{Ker}\phi$ as

$$\{x \in G \mid \phi(x) = \phi(g)\}$$

Now, we can explore the properties of homomorphisms.

Theorem 10.1 (Properties of Elements Under Homomorphisms). Let ϕ be a homomorphism from a group G to a group $\overline{G}$ and let g be an element of G. Then

1. ϕ carries the identity of G to the identity of $\overline{G}$.
2. $\phi(g^n) = (\phi(g))^n$ for all $n \in \mathbb{Z}$.
3. If $|g|$ is finite, then $|\phi(g)|$ divides $|g|$.
4. $\operatorname{Ker}\phi$ is a subgroup of G.
5. $\phi(a) = \phi(b)$ if and only if $a\operatorname{Ker}\phi = b\operatorname{Ker}\phi$.
6. If $\phi(g) = g'$, then $\phi^{-1}(g') = \{x \in G \mid \phi(x) = g'\} = g\operatorname{Ker}\phi$.

Proof.

1. Let $e, \overline{e}$ be the identity elements of G and $\overline{G}$, respectively.

$$\begin{aligned}
\phi(e) &= \phi(e \cdot e) \\
\phi(e) &= \phi(e)\phi(e) \\
\overline{e} &= \overline{e}\phi(e) \\
\overline{e} &= \phi(e).
\end{aligned}$$

2. We have that

$$\begin{aligned}
\phi(g^n) &= \phi(g \cdot g \ldots g) \\
&= \phi(g)\phi(g)\ldots\phi(g) \\
&= (\phi(g))^n
\end{aligned}$$

3. Assume that $|g| = k < \infty$. Then

$$\begin{aligned} \phi(e) &= \phi(g^k) \\ \overline{e} &= \phi(g^k) && \text{(From 1.)} \\ \overline{e} &= (\phi(g))^k && \text{(From 2.)} \end{aligned}$$

Thus, we have that $|\phi(g)|$ divides k.

4. We have that $\operatorname{Ker}\phi \subseteq G$, by definition. We can employ the two-step subgroup test to show it is a subgroup.

   (a) *Closure*: let $a, b \in \operatorname{Ker}\phi$. Thus, $\phi(a) = \overline{e} = \phi(b)$. Then

$$\begin{aligned} \phi(ab) &= \phi(a)\phi(b) \\ &= \overline{e}\,\overline{e} \\ &= \overline{e}. \end{aligned}$$

   Thus, we have $ab \in \operatorname{Ker}\phi$.

   (b) Let $a \in \operatorname{Ker}\phi$. Thus

$$\begin{aligned} \phi(a) &= \overline{e} \\ (\phi(a))^{-1} &= (\overline{e})^{-1} \\ &= \overline{e} \\ &= \phi(a^{-1}). && \text{(By 2.)} \end{aligned}$$

   Thus, $\phi(a^{-1}) = \overline{e}$ and we have $a^{-1} \in \operatorname{Ker}\phi$. By the two-step subgroup test, $\operatorname{Ker}\phi \leq G$.

5. For the forward direction, assume $\phi(a) = \phi(b)$. Let $x \in a\operatorname{Ker}\phi$. Thus $\phi(x) = \phi(a) = \phi(b)$. We have that $x \in b\operatorname{Ker}\phi$, i.e., $a\operatorname{Ker}\phi \subseteq b\operatorname{Ker}\phi$. Similarly, we can prove that $b\operatorname{Ker}\phi \subseteq a\operatorname{Ker}\phi$. Thus, $a\operatorname{Ker}\phi = b\operatorname{Ker}\phi$. For the converse, assume that $a\operatorname{Ker}\phi = b\operatorname{Ker}\phi$. Note that $a \in a\operatorname{Ker}\phi$ and $b \in b\operatorname{Ker}\phi$. Since $a\operatorname{Ker}\phi = b\operatorname{Ker}\phi$, we have that $a \in b\operatorname{Ker}\phi$ and $b \in a\operatorname{Ker}\phi$. Thus, $\phi(a) = \phi(b)$.

6. Assume that $\phi(g) = g'$. We want to show $\phi^{-1}(g') = g\operatorname{Ker}\phi$. We have that

$$x \in \phi^{-1}(g') = \{a \in G \mid \phi(a) = g' = \phi(g)\}.$$

   Thus $\phi(x) = g' = \phi(g)$. We have $x \in g\operatorname{Ker}\phi$. Thus, $\phi^{-1}(g') \subseteq g\operatorname{Ker}\phi$. On the other hand, for $y \in g\operatorname{Ker}\phi$, we have that $\phi(y) = \phi(g) = g'$. We have $y \in \phi^{-1}(g')$. Thus, $g\operatorname{Ker}\phi \subseteq \phi^{-1}(g')$. Thus, we can conclude that $\phi^{-1}(g') = g\operatorname{Ker}\phi$.

□

Since homomorphisms preserve the group operations, it should not be a surprise that they preserve many group properties.

Theorem 10.2 (Properties of Subgroups Under Homomorphisms). Let ϕ be a homomorphism from a group G to a group $\overline{G}$ and let h be a subgroup of G. Then

1. $\phi(H) = \{\phi(h) \mid h \in H\}$ is a subgroup of $\overline{G}$.

2. If H is cyclic, then $\phi(H)$ is cyclic.
3. If H is Abelian, then $\phi(H)$ is Abelian.
4. If H is normal in G, then $\phi(H)$ is normal in $\phi(G)$.
5. If $|\operatorname{Ker}\phi| = n$, then ϕ is an n-to-1 mapping from G onto $\phi(G)$.
6. If $|H| = n$, then $|\phi(H)|$ divides n.
7. If $\overline{K}$ is a subgroup of $\overline{G}$, then $\phi^{-1}(\overline{K}) = \{k \in G \mid \phi(k) \in \overline{K}\}$ is a subgroup of G.
8. If $\overline{K}$ is a normal subgroup of $\overline{G}$, then $\phi^{-1}(\overline{K}) = \{k \in G \mid \phi(k) \in \overline{K}\}$ is a normal subgroup of G.
9. If ϕ is onto and $\operatorname{Ker}\phi = \{e\}$, then ϕ is an isomorphism from G to $\overline{G}$.

Proof.

1. Can be shown using the 2-step subgroup test. Let $x, y \in \phi(H)$. Thus, there are $a, b \in H$ such that $x = \phi(a)$ and $y = \phi(b)$. Therefore

$$\begin{aligned} xy &= \phi(a)\phi(b) \\ &= \phi(ab) \in \phi(H). \end{aligned}$$

   Therefore, we have that $xy \in \phi(H)$. Next, for $x \in \phi(H)$, there is $a \in H$ such that $x = \phi(a)$. Thus, $x^{-1} = (\phi(a))^{-1} = \phi(a^{-1}) \in \phi(H)$. Thus, $x^{-1} \in \phi(H)$, and by the two-step subgroup test, $\phi(H) \leq G$.

2. Let $H = \langle g \rangle$ and $x \in \phi(H)$. There is $a = g^i \in H$ such that $\phi(a) = x$ for some $i \in \mathbb{Z}$. Thus

$$\begin{aligned} x &= \phi(a) \\ &= \phi(g^i) \\ &= (\phi(g))^i. \end{aligned}$$

   We thus have that $\phi(H) = \langle \phi(g) \rangle$.

3. LAETR.
4. Assume that $H \lhd G$. We would like to show that for any $x \in \overline{G}$ that $x\phi(H)x^{-1} \subseteq \phi(H)$.

□

The special case of property 8 of Theorem 10.2, where $\overline{K} = \{e\}$, is quite important, so much so that we will list it as a corollary.

Corollary 19 (Kernels are Normal). Let ϕ be a group homomorphism from G to $\overline{G}$. Then $\operatorname{Ker}\phi$ is a normal subgroup of G.

Proof. Let $\{\overline{e}\}$ be a trivial subgroup of $\overline{G}$. Thus, $\{\overline{e}\} \lhd \overline{G}$, as for any $y \in \overline{G}$

$$y\overline{e}y^{-1} = yy^{-1} = \overline{e} \in \overline{G}.$$

By Theorem 10.2, we have that $\operatorname{Ker}\phi = \phi^{-1}(\{\overline{e}\}) \lhd G$. □

Let's illustrate some properties of Theorems 10.1 and 10.2 with the following examples.

Example 53. Define $\phi : \mathbb{Z}_{12} \to \mathbb{Z}_{12}$ by $\phi(x) = 3x$ for all $x \in \mathbb{Z}_{12}$. We want to show that ϕ is a homomorphism, i.e., ϕ is well defined and ϕ is operation preserving. Let $a, b \in \mathbb{Z}_{12}$

$$\begin{aligned}\phi(a+b) &= 3(a+b)\\ &= 3a+3b\\ &= \phi(a)+\phi(b).\end{aligned}$$

What about $\mathrm{Ker}(\phi)$? Let $x \in \mathrm{Ker}(\phi)$. Thus, $\phi(x) = 3x = 0$, i.e., $12 \mid 3x$. We have that

$$12 \mid 3x \implies 4 \mid x.$$

Therefore, $x = 4k \in \mathbb{Z}_{12}$ for some $k \in \mathbb{Z}$, therefore

$$\mathrm{Ker}(\phi) = \{0, 4, 8\}.$$

Suppose $\overline{K} = \{0, 6\} \lhd \mathbb{Z}_{12}$. We have that

$$\phi^{-1}(\overline{K}) = \{0, 4, 8, 2, 6, 10\} \lhd \mathbb{Z}_{12}\,.$$

Example 54. Determine all homomorphism $\phi : \mathbb{Z}_{12} \to \mathbb{Z}_{30}$. Note that $\phi(x) = \phi(\overbrace{1+1+\cdots+1}^{x \text{ times}})$ for $x \in \mathbb{Z}_{12}$. Without loss of generality, let $x \in \mathbb{N}$. Since we know that

$$\phi(1+1+\cdots+1) = \phi(1)+\cdots+\phi(1),$$

ϕ depends on $\phi(1)$. How many $\phi(1)$'s can we have – i.e., what are the possibilities of the values of $\phi(1)$?

$$\begin{aligned}\phi(x) &= \phi(1+1+\cdots+1)\\ &= \phi(1)+\cdots+\phi(1)\\ &= x\phi(1).\end{aligned}$$

Lagrange's Theorem and property 3 of Theorem 10.1 require that $|\phi(1)|$ divides both 12 and 30. So, we have that $|\phi(1)| = 1, 2, 3$ or 6. Thus, $\phi(1) = 0, 15, 10, 20, 5$, or 25.

10.2 The First Isomorphism Theorem

The next theorem shows that for any homomorphism ϕ of G and the normal subgroup $\mathrm{Ker}\,\phi$, the same process produces a Cayley table isomorphic to the homomorphic image of G.

Theorem 10.1 (First Isomorphism Theorem). Let ϕ be a group homomorphism from G to $\overline{G}$. Then the mapping from $G/\mathrm{Ker}\,\phi$ to $\phi(G)$ given by $g\,\mathrm{Ker}\,\phi \to \phi(g)$, is an isomorphism. In symbols, $G/\mathrm{Ker}\,\phi \cong \phi(G)$.

Proof. Define ψ to denote the correspondence $g\,\mathrm{Ker}\,\phi \to \phi(g)$. The fact that ψ is well-defined and one-to-one follows directly from property 5 of Theorem 10.1. To show that ψ is operation

preserving, observe that

$$\begin{aligned}\psi(x\operatorname{Ker}\phi y\operatorname{Ker}\phi) &= \psi(xy\operatorname{Ker}\phi)\\ &= \phi(xy)\\ &= \phi(x)\phi(y)\\ &= \psi(x\operatorname{Ker}\phi)\psi(y\operatorname{Ker}\phi).\end{aligned}$$

□

The next corollary follows directly from Theorem 10.3, property 1 of Theorem 10.2, and Lagrange's Theorem.

Corollary 20. If ϕ is a homomorphism from a finite group G to $\overline{G}$, then $|\phi(G)|$ divides $|G|$ and $|\overline{G}|$.

Proof. By the First Isomorphism Theorem, $G/\operatorname{Ker}\phi \cong \phi(G)$. Thus,

$$\begin{aligned}|\phi(G)| &= |G/\operatorname{Ker}\phi|\\ &= \frac{|G|}{|\operatorname{Ker}\phi|}.\end{aligned}$$

Therefore, $|\phi(G)|$ divides $|G|$. Since $\phi(G) \leq \overline{G}$, we have that $|\phi(G)|$ divides $|\overline{G}|$. □

Let's now turn our attention to an application of the First Isomorphism Theorem – the N/C Theorem.

Example 55. Let $H \leq G$. Recall that the normalizer of H in G is

$$N(H) = \{x \in G \mid xHx^{-1} = H\}$$

and the centralizer of H in G is

$$C(H) = \{x \in G \mid xhx^{-1} = h, \forall h \in H\}.$$

Note that $C(H) \leq N(H)$. However, do we have that $C(H) \lhd N(H)$. Define $\phi : N(H) \to \operatorname{Aut}(H)$ by $g \mapsto \phi_g$ where $\phi_g(x) = gxg^{-1}$ for each $g \in N(H)$ and $x \in H$. We have that ϕ is a homomorphism. By the First Isomorphism Theorem, we have that $\operatorname{Aut}(H)/\operatorname{Ker}\phi \cong \phi(N(H))$. Furthermore, $\operatorname{Ker}\phi = C(H)$.

Chapter 11

Fundamental Theorem of Finite Abelian Groups

11.1 The Fundamental Theorem

In this chapter, we will take a look at a theorem that describes all finite Abelian groups in a standardized way.

Theorem 11.1 (Fundamental Theorem of Finite Abelian Groups). Every finite Abelian group is a direct product of cyclic groups of prime-power order. Moreover, the number of terms in the product and the others of the cyclic groups are uniquely determined by the group.

This reduces questions about finite Abelian groups to questions about cyclic groups, which can generally be answered using Chapter 4. Since a cyclic group of order n is isomorphic to $\mathbb{Z}_n$, Theorem 11.1 shows that every finite Abelian group G is isomorphic to a group of the form

$$\mathbb{Z}_{n_1} \times \mathbb{Z}_{n_2} \times \cdots \times \mathbb{Z}_{n_k},$$

for some $n_1, n_2, \ldots, n_k \in \mathbb{Z}$ such that $n_1 \cdot n_2 \ldots n_k = n$. We will not be proving this, as the proof is quite long and difficult.

Chapter 12

Introduction to Rings

12.1 Motivation and Definition

Many sets are paired with two binary operations: addition and multiplication. For example, integers, real numbers, matrices, and polynomials. Previously, we simply used addition and ignored multiplication. However, in my cases, we want to take multiplication into account as well. The concept of a *ring*[*] does just that.

Definition (Ring). A ring R is a set with two binary operations, addition (denoted by $a+b$) and multiplication (denoted by ab) such that for all a,b,c in R:

1. $a+b=b+a$
2. $(a+b)+c=a+(b+c)$
3. $\exists 0 \in R$ s.t. $a+0=a, \forall a \in R$
4. $\exists -a \in R$ s.t. $a+(-a)=0$
5. $a(bc)=(ab)c$
6. $a(b+c)=ab+ac$ and $(b+c)a=ba+ca$.

In other terms, a ring is an Abelian group under addition, also having associative multiplication that is left and right distributive over addition. We do note a couple things

- Multiplication does not need to be commutative. When it is, we say that the ring is *commutative*.
- A ring does not need an identity under multiplication. A *unity* (or *identity*) in a ring is a nonzero element that is an identity under multiplication.
- A nonzero element of a commutative ring with unity need not have a multiplicative inverse. When it does, we say that it is a *unit* of the ring. In other terms, a is a unit if a^{-1} exists.

Due to this, we call $(R,\cdot)$ a *semi-group*.

[*]Originated in the mid-19th century by Richard Dedekind, but its first formal abstract definition given by Abraham Fraenkel in 1914. The term *ring* was first applied in 1897 by Hilbert.

Remark 10. Sets that have more than two operations, e.g., $(A, +, \cdot, *, \star, \ldots)$, are studied in a field called *universal algebra*.

12.2 Examples of Rings

The following are some examples of rings.

Example 56. The set $\mathbb{Z}$ of integers under ordinary addition and multiplication is a commutative ring with unity 1. The units of $\mathbb{Z}$ are 1 and -1.

Example 57. The set $\mathbb{Z}_n = \{0, 1, \ldots, n-1\}$ under addition and multiplication modulo n is a commutative ring with unity 1. The set of units is $U(n)$.

Example 58. The set $\mathbb{Z}[x]$ of all polynomials in the variable x with integer coefficients, i.e.,

$$\mathbb{Z}_{[x]} = \{a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x_1 \mid a_i \in \mathbb{Z}, \forall i = 1, 2, \ldots, n\}$$

under ordinary addition and multiplication is a commutative ring with unity $f(x) = 1$.

Example 59. The set $M_2(\mathbb{Z})$ of 2×2 matrices with integer entries is a non-commutative ring with unity $\left(\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}\right)$.

12.3 Properties of Rings

Next we will show how the operations of addition and multiplication interact. Note that we will not refer to $a - b$ as $a + (-b)$.

Theorem 12.1 (Rules of Multiplication). Let a, b, and c belong to a ring R. Then

1. $a0 = 0a = 0$
2. $a(-b) = (-a)b = -(ab)$
3. $(-a)(-b) = ab$
4. $a(b - c) = ab - ac$ and $(b - c)a = ba - ca$

Furthermore, if R has a unity element 1, then

5. $(-1)a = -a$
6. $(-1)(-1) = 1$

Proof.

1. We can use the fact that $0 = 0 + 0$ and the distributive property.

$$\begin{aligned}
0 &= 0 + 0 && \text{(additive identity)}\\
a0 &= a(0+0) \\
a0 &= a0 + a0 && \text{(distribution)}\\
a0 + (-(a0)) &= (a0 + a0) + (-(a0)) && \text{(additive inverse)}\\
0 &= a0 + (a0 + (-(a0))) && \text{(associativity)}\\
0 &= a0
\end{aligned}$$

   therefore, we must have that $a0 = 0$. The proof for the mirror case $0a = 0$ can be shown similarly.

2. We can use property 1

$$\begin{aligned}
0 &= a0 && \text{(property 1)}\\
0 &= a(b-b) \\
0 &= ab + a(-b) && \text{(distributive)}\\
0 + (-(ab)) &= (ab + a(-b)) + (-(ab)) \\
-(ab) &= a(-b) + (ab + (-(ab))) && \text{(associativity)}\\
-(ab) &= a(-b) && \text{(additive inverse)}
\end{aligned}$$

   The mirror case can be shown similarly.

3. Use property 2.

4. LAETR.

5. Assume that R has a unity element 1

$$\begin{aligned}
0 &= (1 + (-1)) \\
&= 1a + (-1)a \\
&= a + (-1)a \\
0 + (-a) &= (a + (-1)a) + (-a) \\
&= (-1)a + (a + (-a)) \\
-a &= (-1)a
\end{aligned}$$

6. We want to show that $(-1)(-1) = 1$. By property 3, we have that

$$\begin{aligned}
(-1)(-1) &= 1 \cdot 1 \\
&= 1.
\end{aligned}$$

□

Recall that in the case of groups, the identity and inverses are unique. The same is true for rings, provided that these elements exist.

Theorem 12.2 (Uniqueness of the Unity and Inverses). If a ring has unity, it is unique. If a ring element has a multiplicative inverse, it is unique.

Proof. LAETR – think about the definition of a group. □

12.4 Subrings

While studying groups, subgroups played a crucial role. The analogous structures in ring theory, *subrings*, play a less important rule but are still worth studying.

Definition (Subring). A set S of a ring R is a subring of R if S is itself a ring with the same operations of R.

Example 60. $\{0\}, R$ are trivial subrings of a ring R.

Example 61. $(\mathbb{Z}, +, \cdot)$ is a subring of $(\mathbb{R}, +, \cdot)$.

Similarly as there was for subgroups, there is a simple test for subrings.

Theorem 12.3 (Subring Test). A nonempty subset S of a ring R is a subring if S is closed under subtraction and multiplication – that is, if $a - b$ and ab are in S whenever $a, b \in S$.

Chapter 13

Integral Domains

13.1 Definition and Examples

The notion of a ring was invented in an attempt to put the algebraic properties of integers into an abstract setting. However, we saw that rings lose some properties of the integers – commutativity and existence of unity, and also the cancellation property. There are a particular class of rings that actually have all three of these: *integral domains*. First, we introduce the concept of *zero-divisors*.

Definition (Zero-Divisors). A *zero-divisor* is a nonzero element a of a commutative ring R such that there is a nonzero element $b \in R$ with $ab = 0$ or $ba = 0$.

Example 62. $M_2(\mathbb{R})$ is a ring

$$\begin{aligned} a &= \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \neq 0 \\ b &= \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \neq 0 \\ ab &= \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \\ &= \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \end{aligned}$$

Therefore, a is a zero divisor of $M_2(\mathbb{R})$.

Now we can introduce the definition of integral domains.

Definition (Integral Domain). An *integral domain* is a commutative ring with unity and no zero-divisors.

Thus, in an integral domain, a product is 0 only when one of the factors is 0; that is, $ab = 0$ only when $a = 0$ or $b = 0$.

Example 63. The ring of integers is an integral domain.

Example 64. The ring of Gaussian integers $\mathbb{Z}[i] = \{a + bi \mid a, b \in \mathbb{Z}\}$ is an integral domain.

Example 65. The ring $\mathbb{Z}[x]$ of polynomials with integer coefficients is an integral domain.

Example 66. The ring $\mathbb{Z}_n$ of integers modulo n is an integral domain when n is prime.

Example 67. The ring $M_2(\mathbb{Z})$ of 2×2 matrices over the integers is *not* an integral domain

What makes integral domains special is that they have cancellation, in spite of the fact that the nonzero elements need not form a group under multiplication.

Theorem 13.1 (Cancellation). Let a, b, and c belong to an integral domain. If $a \neq 0$ and $ab = ac$, then $b = c$.

Proof. From $ab = ac$, we have that $a(b - c) = 0$. Since $a \neq 0$, we must have $b - c = 0$. □

Integral domains can also be defined by the cancellation property – that is, as commutative rings with unity in which the cancellation property holds. The definition is equivalent to the one we have.

13.2 Fields

In many applications, a particular kind of integral domain called a *field* is necessary.

Definition (Field). A *field* is a commutative ring with unity in which every nonzero element is a unit.

Every field is an integral domain as if a and b belong to a field with $a \neq 0$ and $ab = 0$, we can multiply both sides of the last expression by a^{-1} to obtain $b = 0$. Note that $(F, +, \cdot)$ is a commutative ring such that

1. $(F, +)$ is an Abelian group
2. $(F \setminus \{0\}, \cdot)$ is an Abelian group

Example 68. $(\mathbb{Q}, +, \cdot)$ is a field.

Example 69. $(\mathbb{R}, +, \cdot)$, $(\mathbb{C}, +, \cdot)$ are fields.

The next result states that in the finite case, fields and integral domains are the same.

Theorem 13.2 (Finite Integral Domains Are Fields). A finite integral domain is a field.

Proof. Let R be a finite integral domain. Let $a \in R \setminus \{0\}$. We want to show that a has a

multiplicative inverse. If $a = 1$, we are done. Assume $a \neq 1$. Consider $\{a, a^2, a^3, \dots\} \subseteq R$. Since R is finite, $\{a, a^2, a^3, \dots\}$ is finite. Thus, there are $i, j \in \mathbb{N}$ such that $i < j$ and $a^i = a^j$. Therefore, we have that

$$a^{j-i} = 1 \iff a^{j-i-1} \cdot a = 1.$$

Thus a has a^{j-i-1} as a multiplicative inverse. □

One quick corollary* of this theorem is the following

Corollary 21 ($\mathbb{Z}_p$ Is a Field). For every prime p, $\mathbb{Z}_p$, the ring of integers modulo p is a field.

*And the course's final corollary :(